

Energy Conditions and Weakening the Cosmic No-Hair Theorem

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Abstract

In this thesis, we study various types of energy conditions used in general relativity and quantum field theory, particularly in the context of the Cosmic No-Hair Theorem. We first review pointwise energy conditions and derive the Raychaudhuri and Gauss-Codazzi equations. Next, we review Wald's Cosmic No-Hair Theorem, which considers initially expanding homogeneous spacetimes satisfying Einstein's equation with positive cosmological constant Λ . This theorem shows that, for all cosmological models except Bianchi type IX, the spacetime locally approaches de Sitter at late times. We then review averaged energy conditions and quantum energy inequalities, which are generally weaker than their pointwise counterparts. We conclude by discussing the effects of using some of these weaker energy conditions in the framework of the Cosmic No-Hair Theorem.

Contents

1	Introduction	4
2	Pointwise energy conditions	6
3	Background for the Cosmic No-Hair Theorem	9
3.1	Homogeneous spacetimes and the 1+3 approach	9
3.2	The Raychaudhuri Equation	11
3.3	The Gauss-Codazzi Equations	14
3.4	The Bianchi Classification	16
4	The Cosmic No-Hair Theorem	18
5	Averaged energy conditions	19
6	The non-minimally coupled scalar field	21
6.1	A strong worldline energy inequality	22
6.2	A weak worldline inequality	24
7	Quantum Energy Inequalities	25
7.1	QFT violating energy conditions - example	26
7.2	Klein-Gordon field QEI	27
8	Weakening energy conditions in the Cosmic No-Hair Theorem	27
9	Conclusion	30

1 Introduction

Our modern understanding of the universe is underpinned by Einstein’s theory of general relativity, which asserts that the concept of gravity is a consequence of the geometrical nature of spacetime. This is achieved through Einstein’s field equations, which relate the physical properties of matter to the geometrical properties of spacetime which it inhabits. One immediate difficulty that arises from this formulation is that, unlike some other physical theories where a dynamical variable evolves on a fixed background, here, the metric and matter inside the spacetime are self-referential, meaning one has to solve for both at the same time. This property of space and matter makes Einstein’s equations extremely difficult to solve, and one usually has to impose some sort of additional constraints or symmetries on the spacetime in order to make progress. For example, one could demand spatial symmetry or a matterless universe. At first glance doing this may seem restrictive, however these constraints are the reason why many solutions to Einstein’s equation are discovered. Schwarzschild discovered his eponymous solution by searching for a spatially symmetric and vacuum solution to the Einstein equations [1]. Another reason why restricting the matter in a spacetime may be important, is that too liberal of a constraint may give rise to unphysical phenomena like wormholes, naked singularities or closed timelike curves, the likes of which have never been observed in our universe. From these points, a question arises: how do we restrain the matter in our universe in a way that is restrictive enough to not allow unreasonable physical phenomena, but relaxed enough to still give solutions to Einstein’s equation?

The concept that attempts to answer this question is that of *energy conditions*, which are pointwise conditions placed on the stress tensor of a theory which aim to provide a physically reasonable description of how matter and energy behave. They were first used in the context of proving singularity theorems by Penrose and Hawking [2, 3], in which Hawking states “These are properties that any normal matter should have”. These properties in question include notions like “energy should be non-negative” and “gravity should be attractive”, which are conceptually simple and intuitive and have contributed to the popularity of energy conditions since their inception. The practicality of energy conditions is further improved by the fact that through analysing the properties of matter considering only its stress tensor, precise details of the matter need not be known. Not requiring knowledge of the complete internal structure of the stress tensor enables a variety of theorems to be proven, often in a relatively streamlined manner. Singularity theorems [2, 4, 5], Black Hole Uniqueness Theorems [6], the Black Hole Area Theorem [7], The Cosmic No-Hair Theorem [8] and other assorted theorems [9, 10] all depend on pointwise energy conditions for their results, and have greatly contributed to the development of general relativity in the latter half of the twentieth century.

Despite their promise, pointwise energy conditions are not infallible. One reason for this is that currently, there is no consensus on a preferred energy condition - we have a variety of energy conditions that are used in different contexts and have different strengths, weaknesses, and ranges of validity. This is partly because, as discussed in [11], we have no clear understanding of the exact nature of these conditions or how they might arise in fundamental physics. This makes it difficult to determine the types of situations where they may appear, and gives us no way to interpret their consequences. Ultimately, it is physicists who create the energy conditions, and there should be a balance to devising conditions strong enough to help us prove results, whilst also being physically reasonable. When this balance shifts, it can be interpreted as authors reverse engineering their theorems to find energy conditions that are tailored to their specific needs [12]. The validity of pointwise energy conditions is also seriously questioned as they can be violated by simple classical and quantum fields, and further, they can be violated arbitrarily badly [13, 14, 15].

In response to this problem, *averaged energy conditions* were created, which place conditions on the stress tensor over a spacetime region, usually a causal geodesic. One interpretation of these averaged

conditions is that the violation of pointwise energy conditions on small scales is allowed, but not enough to have a considerable impact over a large region. These averaged conditions are weaker than their pointwise counterparts and are harder to violate. The non-minimally coupled scalar field easily violates all four of the main pointwise energy conditions, and while violating the averaged conditions is still possible, it is generally more difficult. For example, when considering the non-minimally coupled scalar field, the *averaged null energy condition* (ANEC) can be violated only when the scalar field takes trans-Planckian values [16]. Hence, due to the logical implications between energy conditions, as we will discuss, all the average energy conditions are violated in this situation. The *achronal null energy condition* (AANEC), which is the same as the ANEC with the additional requirement that the average is taken over achronal curves (meaning no two points on the curve can be connected by a timelike curve), is the mostly likely candidate of the averaged energy conditions to hold generally. Should the AANEC hold, there are theorems within semiclassical gravity that show the positivity of energy, the topological censorship theorem, and the impossibility of closed timelike curves [17].

Considering the apparent advantages of using averaged energy conditions instead of pointwise ones, one can ask whether certain results that depend on pointwise energy conditions can also be proven solely with averaged conditions, or at least, one can consider the effects on a certain result when pointwise conditions are replaced with averaged ones. Recently, there have been works which establish analogues of the Penrose-Hawking singularity theorems and the Hawking Black Hole Area Theorem using average conditions, replacing the pointwise conditions used in their original formulations [18, 19].

The most prominent article we look at in this thesis is Wald’s 1983 paper [8], which asserts that an initially expanding homogenous spacetime under certain pointwise energy conditions will asymptotically approach de Sitter spacetime. Throughout, we call this result the “Cosmic No-Hair Theorem”. De Sitter space is the counterpart of the three-sphere of Euclidean geometry and is one of the three maximally symmetric solutions to the vacuum Einstein equations. These solutions are characterised by the sign of the cosmological constant Λ . Minkowski space corresponds to $\Lambda = 0$, whilst de Sitter and anti-de Sitter space correspond to $\Lambda > 0$ and $\Lambda < 0$ respectively. Arguably, de Sitter space is of the most physical interest, as astronomical observations have shown that our universe does indeed have a positive, albeit small, cosmological constant [20, 21].

In this thesis, we investigate the effects of weakening the pointwise energy conditions used in the Cosmic No-Hair Theorem, described above. The motivation for doing this, as well as the discussion about energy conditions above, is that there is recent numerical evidence to suggest that the Cosmic No-Hair Theorem does hold under energy conditions that are weaker than the original pointwise ones [22, 23]. We will see in Sec. 8 that under certain averaged conditions, whilst still keeping some pointwise ones, we can obtain the result that under many of the same assumptions as the original Cosmic No-Hair Theorem, the universe will approach de Sitter spacetime. We will see, however, that there are certain trade-offs between obtaining the results one wants, versus the amount of conditions it is necessary to assume, and this invites a debate of how many requirements one can place on the matter and geometry of a spacetime without it becoming physically unreasonable. As we will see in Sec. 8, there is still plenty of room for more discussion in regards to weakening the Cosmic No-Hair Theorem, since the weakened energy conditions and techniques used are in no sense unique, and so the possibility of finding stronger results remains.

Conventions and notation

We use units $c = G = \hbar = 1$. We denote a spacetime by (M, g_{ab}) which we take to be time orientable and use metric signature $(-+++)$. We use ∇ to denote the Levi-Civita connection for the metric g_{ab} . The Riemann tensor is defined as $R_{abc}{}^d \omega_d = \nabla_a \nabla_b \omega_c - \nabla_b \nabla_a \omega_c$. The Ricci tensor is then defined as $R_{ab} = R_{acb}{}^c$ and the Riemann curvature as $R = R_{ab} g^{ab}$. We use the abstract tensor notation from [24].

2 Pointwise energy conditions

In the broadest sense, an *energy condition* is a condition imposed on the stress-energy tensor (often just called the stress tensor) of matter which aims to quantify reasonable physical notions about how spacetimes behave. This includes ideas like the “positivity of energy” or the “attractiveness of gravity” although as we will see specifics of these ideas contain plenty of room for discussion. Classical energy conditions are often grouped into two categories: pointwise energy conditions, which are conditions that are demanded to hold at each point spacetime, and average conditions, which are conditions that are demanded to hold when averaged over some spacetime region. In this section, we give an overview of the most common pointwise energy conditions, and we give a discussion of their interpretations. Later in Sec. 5, we will discuss averaged energy conditions, which are often considered to be the weakened counterparts of the pointwise energy conditions.

Within the class of pointwise energy conditions, we have a choice of how each individual energy condition is formulated. These formulations are given the names *physical*, *geometric*, and *effective*; the formulation we choose will depend on the specific situation we are working in. Physical energy conditions are usually the most intuitive and are often the way in which energy conditions are stated. In this formulation, conditions are placed on the stress-energy tensor T_{ab} and so directly restrict the form of the matter in the spacetime. Geometric energy conditions are placed on a purely geometric tensor, usually the Ricci or Einstein tensor, and we can think of these conditions as restricting the geometry of a spacetime due to the matter inside it. If the Einstein equation is assumed to hold then physical and geometric energy conditions are equivalent and one uses the Einstein equation to go between the two [11]. We will see precisely how with is done in Sec. 3.2, where we will also make the important point that different physical and geometric formulations will be found depending on whether one includes a cosmological constant in the Einstein equation or not. To state the effective formulation, we first note from [25] that for “reasonably behaved” matter, the stress-energy tensor can be written in matrix form as

$$T^{ab} = \text{diag}(\rho, P_1, P_2, P_3), \quad (2.1)$$

or equivalently in tensor form as

$$T^{ab} = \rho t^a t^b + P_1 x^a x^b + P_2 y^a y^b + P_3 z^a z^b, \quad (2.2)$$

where $\{t^a, x^a, y^a, z^a\}$ is an orthonormal basis with t^a timelike and x^a, y^a, z^a spacelike. Here, we are using four spacetime dimensions but this can be extended to any number of spacetime dimensions in the obvious way. For this case, in which the matter is sometimes called a “relativistic fluid”, ρ is interpreted as the energy density as measured by an observer travelling on a world line with tangent vector t^a , while the P_1, P_2, P_3 represent the principle pressures in the spacetime directions x^a, y^a, z^a respectively. An effective energy condition is then a condition placed on ρ and P_i . As noted in [25], all experimentally detected fields have stress tensors that can be written in this form so it is reasonable to include this formulation in our description of pointwise energy conditions. All three descriptions are equivalent if Einstein’s equation is assumed to hold and the matter is reasonably well-behaved so that it has a stress tensor that can be written in the form of Eq. (2.1). Below, we list the most commonly used pointwise energy conditions in their physical, geometric, and effective formulations in four spacetime dimensions.

Weak energy condition (WEC)

physical for all timelike vectors n^a , $T_{ab}n^a n^b \geq 0$

geometric for all timelike vectors n^a , $G_{ab}n^a n^b \geq 0$

effective $\rho \geq 0$ and for all i , $\rho + P_i \geq 0$

Strong energy condition (SEC)

physical for all timelike vectors n^a , $(T_{ab} - \frac{1}{2}Tg_{ab})n^an^b \geq 0$

geometric for all timelike vectors n^a , $R_{ab}n^an^b \geq 0$

effective $\rho + \sum_i P_i \geq 0$ and for all i , $\rho + P_i \geq 0$

Null energy condition (NEC)

physical for all null vectors k^a , $T_{ab}k^ak^b \geq 0$

geometric for all null vectors k^a , $R_{ab}k^ak^b \geq 0$

effective for all i , $\rho + P_i \geq 0$

Dominant energy condition (DEC)

physical for all co-oriented timelike vectors m^a and n^a , $T_{ab}m^an^b \geq 0$

geometric for all co-oriented timelike vectors m^a and n^a , $G_{ab}m^an^b \geq 0$

effective $\rho \geq 0$ and for all i , $\rho \geq |P_i|$

Despite its name, the SEC does not imply the WEC, however the energy conditions do obey the hierarchy of implications seen in Fig. 1. These implications are easily derived as follows. As the three formulations are equivalent we can focus on the physical formulation. We can see that the DEC implies the WEC by taking $m^a = n^a$, the WEC implies the NEC by continuity, and the SEC implies the NEC by continuity since $g_{ab}n^an^b \rightarrow 0$ after taking n^a to approach a null vector.

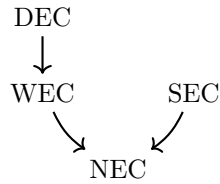


Figure 1: Implications of energy conditions where an arrow denotes implication.

We now discuss the interpretations of the pointwise energy conditions stated above. There is not a clear interpretation of every energy condition in every formulation, however the interpretations we do provide can give useful insight and intuition about the restrictions they place on spacetime. Many of the following interpretations come from [11], and one can refer to this review for more discussion.

The interpretation of the physical form of the WEC is the most intuitive, it is that the energy density recorded by an observer travelling on a timelike curve is non-negative, where the energy density ρ is defined as

$$\rho = T_{ab}n^an^b, \quad (2.3)$$

and n^a is the timelike tangent vector of the traveller's trajectory. This is one of the most intuitive interpretations as it makes precise the idea that “energy should be positive” which we mentioned earlier. The geometric form of the WEC does not have an interpretation as simple, although we do have the following description from in [26]. We let $V > 0$ be the area of a small spacelike geodesic ball, and consider the ratio of this spatial volume with the spatial volume of the corresponding ball in Minkowski space. Then the geometric WEC holds if and only if this area is greater or equal to 1 in the limit when the ball shrinks to a point. The interpretation of the effective form is relatively intuitive. It demands that the energy density measured along a timelike curve is non-negative, and in addition, the pressure (which we can think of as a sort of relativistic mass-energy flow) in any spacetime direction is allowed to be negative, but cannot be so negative that it overcomes the energy density.

There is little interpretation of the physical form of the SEC apart from the idea of merely requiring the *effective energy density* (EED) measured along a timelike curve to be non-negative. The EED is often also denoted by ρ and is defined as

$$\rho = \left(T_{ab} - \frac{1}{2} T g_{ab} \right) n^a n^b, \quad (2.4)$$

where again n^a is the timelike tangent vector of the traveller's trajectory. There is no real physical argument as to why the EED should hold instead of the energy density seen in the WEC. For a geometric interpretation of the SEC, we use an argument from [11] which involves idea of average radial acceleration. From the geodesic deviation equation, we can obtain an expression for the *average radial acceleration* A_r ,

$$A_r = -\frac{1}{3} R_{ab} n^a n^b, \quad (2.5)$$

where n^a as usual is the timelike tangent vector of the traveller's trajectory. From Eq. (2.5), we can see that the geometric SEC holds if and only if $A_r \leq 0$. This means that nearby timelike geodesics will converge (or at least not diverge) to one another in sufficiently small neighbourhoods of every spacetime point. Intuitively, this means that observers on timelike curves feel an “attractive force of gravity”. The effective form of the SEC has the same kind of interpretation as the WEC. This is an observer travelling on a timelike curve can measure the energy density and pressure in any spacelike direction both as negative, but their sum must remain non-negative, i.e., the pressure cannot overcome the energy density. In addition to this, the sum of the principal pressures cannot be so negative as to “dominate” the energy density. There is little explanation for the physical meaning of this formulation.

For the physical interpretation of the NEC, one can either say that, in a similar way to the WEC, that the energy density measured by an observer along a null geodesic is non-negative, or one can use again the idea of average radial acceleration. For null geodesic, we can use the geodesic deviation to obtain the expression for A_r

$$A_r = -4\pi T_{ab} k^a k^b, \quad (2.6)$$

where k^a is the null tangent vector of the traveller's trajectory. So the physical NEC holds if and only if $A_r \leq 0$, so null geodesics tend to converge (or at least not diverge) locally around spacetime points, which again implies that they feel a sort of “attractive force of gravity”. For the geometric form, one can obtain a very similar equation to that of Eq. (2.5) and so can apply the same arguments as above. The effective interpretation of the NEC, is that an observer travelling on a null curve can measure both energy density and principal pressure to be negative, but their sum total must remain non-negative. Another interpretation, similar to the WEC, is that an observer travelling on a like curve measures the energy density to be non-negative.

The physical form of the DEC says that an observer travelling on a timelike curve measures the energy density to be non-negative, and also measures the flux of energy to be casual when this flow and the observer's direction are co-oriented. The geometric form of the DEC, like the WEC, does not have a

readily intuitive interpretation. The effective interpretation of the DEC is that firstly, the energy density as measured by an observer on a timelike curve is non-negative, and secondly that the principal pressures are bounded above and below by the energy density. In a sense, this means that the relativistic fluid is not too ‘rigid’ or too ‘thin’ and sits in an ideal zone.

Now with some intuition for the main pointwise energy conditions, a natural question to ask is whether the lower bound of zero in the statements of the conditions can be lowered. In [27], it was first shown that this isn’t possible for the WEC, and further, in [26] it is shown that this isn’t possible for any of the pointwise energy conditions stated above. Specifically, [26] shows that if any of the energy conditions above are bounded below, then this bound is necessarily zero. In a certain sense, this result confirms to us that the pointwise energy conditions above are well-defined and meaningful, even with their limitations.

3 Background for the Cosmic No-Hair Theorem

The *Cosmic No-Hair Theorem* is the name given to the theorem proven by Wald in the 1983 paper [8], and is one of the first instances of using the pointwise energy conditions described above to obtain results about the expansion and asymptotic behaviour of the universe. Before we prove the Cosmic No-Hair Theorem in Sec. 4, we need to develop some machinery that will help us navigate [8] and will help us describe the spacetimes we encounter. We first define *homogeneous* and *foliated* spacetimes describe what are called “1+3 approaches”, which we will see are ways of describing a spacetime as an evolution of “space” in time. Next, we will derive the *Raychaudhuri equation*, which relates the motion of a geodesic to the geometry of the spacetime. Afterwards, we derive one of the *Gauss-Codazzi equations*, which relates the geometry of the full spacetime to the geometry of the “space” moving through time. Finally, we look at the *Bianchi classification*, which is a convenient way of listing the types of spacetimes that are possible in our framework.

3.1 Homogeneous spacetimes and the 1+3 approach

One important assumption that we make about our spacetime going forward, is that it is *homogeneous*. Intuitively, this means that at any point in time, no point in space is preferred over any other point. To formulate this precisely, it is first useful to give a definition of a *foliated spacetime* found in [28].

Definition 1. A spacetime (M, g_{ab}) is said to be *foliated* by spacetime hypersurfaces if there exists a smooth scalar field $\hat{t} : M \rightarrow \mathbb{R}$ with non-vanishing gradient such that each hypersurface is a level set of this scalar field,

$$\forall t \in \mathbb{R}, \quad \Sigma_t = \{p \in M \mid \hat{t}(p) = t\}, \quad (3.1)$$

and such that these hypersurfaces are spacelike and cover M ,

$$M = \bigcup_{t \in \mathbb{R}} \Sigma_t. \quad (3.2)$$

Since $\hat{t}$ has non-vanishing gradient, we have $\Sigma_t \cap \Sigma_{t'} = \emptyset$ for $t \neq t'$, i.e. Eq. (3.2) is a disjoint union. From this point forward, a hypersurface will always denote a *spacelike* hypersurface (i.e. all tangent vectors to the hypersurface are spacelike).

Intuitively, we can think of t , which we can consider as being equivalent to $\hat{t}$, as a “time function” in which each $t \in \mathbb{R}$ specifies a unique point in time given by the hypersurface Σ_t which “fills” the spacetime as t ranges through $\mathbb{R}$. This will be useful later as taking $t \rightarrow \infty$ will give us the asymptotic behaviour of the spacetime. Further, the gradient of t , which has components $\nabla^a t = g^{ab} \nabla_b t$, is a timelike vector

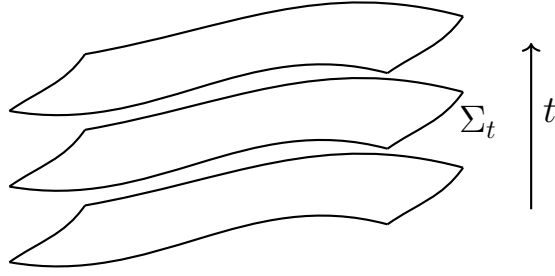


Figure 2: A spacetime foliated by spacelike hypersurfaces Σ_t with a time direction indicated by an arrow.

normal to the hypersurfaces and gives a unique (up to a sign) normalised timelike vector field which is normal to the hypersurfaces by

$$n^a = [-(\nabla^b t)(\nabla_b t)]^{-1/2} \nabla^a t. \quad (3.3)$$

This technique of splitting up a spacetime into spacelike “slices” that move through time is often called the “1+3 approach”, and one can refer to [28] for further reading. Now, we can provide the definition of a homogeneous spacetime, for which we use a definition from [24].

Definition 2. A spacetime (M, g_{ab}) is said to be (spatially) *homogeneous* if it can be foliated by spacelike hypersurfaces $\{\Sigma_t\}_{t \in \mathbb{R}}$ such that for each t and for any points $p, q \in \Sigma_t$ there exists an isometry of the spacetime metric, g_{ab} , that takes p to q . Here an *isometry* of g_{ab} from p to q is a diffeomorphism $\phi : M \rightarrow M$ such that $(\phi^* g)_{ab} = g_{ab}$ (where ϕ^* denotes the pullback of ϕ) and $\phi(p) = q$. See Appendix C in [24] for a fully precise definition of an isometry.

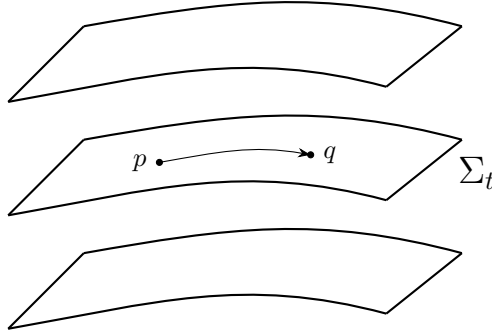


Figure 3: A spatially homogenous spacetime. Here there is an isometry from p to q on the hypersurface Σ_t .

With these definitions in hand, the next step is to derive relationships between geometrical quantities on the full spacetime and the corresponding quantities on the hypersurfaces. The motivation for this is that the asymptotic behaviour of the hypersurfaces will in turn tell us what sort of universe we end up with at these late times. The key first step is to notice that the spacetime metric g_{ab} induces a *spatial metric* (a three-dimensional Riemannian metric) h_{ab} on each of the hypersurfaces Σ_t by the formula

$$h_{ab} = g_{ab} + n_a n_b \quad (3.4)$$

where n^a is the normal to the hypersurfaces defined by Eq. (3.3). This spatial metric h_{ab} gives rise to two particularly useful quantities, namely the *extrinsic curvature*, K_{ab} , and the *spatial curvature*, ${}^{(3)}R$, where both quantities are understood to be evaluated on the hypersurfaces Σ_t . The following two subsections

derive the *Raychaudhuri equation* and one of the *Gauss-Codazzi* equations, which are two equations used to obtain the results in [8].

3.2 The Raychaudhuri Equation

To begin, the derivation of the Raychaudhuri equation, we first define the extrinsic curvature of the hypersurfaces K_{ab} as

$$K_{ab} = \frac{1}{2} \mathcal{L}_n h_{ab}, \quad (3.5)$$

where $\mathcal{L}_n$ denotes the Lie derivative with respect to the vector field n^a . Since we think of n^a as the vector pointing towards increasing time, K_{ab} measures how much the hypersurfaces curves as time progresses and so we also have that

$$\mathcal{L}_n h_{ab} = \dot{h}_{ab}, \quad (3.6)$$

i.e. the derivative of h_{ab} with respect to proper time. Eq. (3.6) shows that K_{ab} is symmetric and further we have the calculation,

$$\begin{aligned} K_{ab} &= \frac{1}{2} \mathcal{L}_n h_{ab} = \frac{1}{2} \mathcal{L}_n (g_{ab} + n_a n_b) \\ &= \frac{1}{2} (n^c \nabla_c g_{ab} + g_{cb} \nabla_a n^c + g_{ac} \nabla_b n^c) = \frac{1}{2} (\nabla_a n_b + \nabla_b n_a). \end{aligned} \quad (3.7)$$

In the second equality above we used the definition of h_{ab} from Eq. (3.4) and in the third equality above we use Eq. (C.2.14) from [24], which is a formula for the Lie derivative of an arbitrary tensor along a vector field, and also the fact that a Lie derivative of a vector field along itself vanishes. The result is further confirmation that K_{ab} is symmetric and also motivates the definition of the tensor B_{ab} as

$$B_{ab} = \nabla_b n_a, \quad (3.8)$$

so that

$$K_{ab} = B_{(ab)}. \quad (3.9)$$

Next, we will give another definition from [24].

Definition 3. Let M be a manifold and $O \subset M$ be open. A *congruence* in O is a family of curves such that for each point $p \in M$ exactly one curve in this family passes through p . The tangents to a congruence yield a vector field in O , and conversely, a vector field in O gives rise to a congruence.

In our case, we will consider smooth and timelike geodesic congruences, that is the congruence will be generated by the timelike vector field n^a , and we assume the integral curves are smooth geodesics (they will also be timelike by construction). Shortly, we will define the terms *expansion*, *shear* and *twist* of a geodesic congruence which will then be related to the geometry of the spacetime. To begin this process, we note that the tensor B_{ab} is purely spatial, meaning

$$B_{ab} n^a = B_{ab} n^b = 0. \quad (3.10)$$

The first equality is shown by the calculation

$$B_{ab} n^a = (\nabla_b n_a) n^a = \nabla_b (n_a n^a) - (\nabla_b n^a) n_a = -B_{ab} n^a, \quad (3.11)$$

where we have used the fact that n^a is normalised and have also simultaneously raised and lowered the index a . The second equality is shown by the calculation

$$B_{ab} n^b = (\nabla_b n_a) n^b = g_{ac} (\nabla_b n^c) n^b = g_{ac} (\partial_b n^c + \Gamma_{bd}^c n^d) n^b = g_{ac} (\dot{n}^c + \Gamma_{bd}^c n^b n^d) = 0, \quad (3.12)$$

where the last equality holds since n^a is the tangent of a geodesic. This is also just the statement that the parallel transport of n^a along itself is zero. Next, for some geometric intuition, we consider a one parameter family $\gamma_s(\tau)$ in the congruence, i.e., for each s , γ_s is a geodesic in the congruence parametrised by τ . Then, we define an orthogonal *deviation vector* η^a (see [24] for details) which, through reparametrisation we can always choose to be normalised and orthogonal to n^a . With this parametrisation, $n^a = (\partial_\tau)^a$ and $\eta^a = (\partial_s)^a$ provide a coordinate system (τ, s) on the surface Σ spanned by the one parameter family $\gamma_s(\tau)$, and hence their Lie bracket vanishes. This is due to the fact that we have constructed an orthonormal basis of the tangent space at each point in Σ and so the partial derivatives commute. Hence, now we have that

$$0 = \mathcal{L}_n \eta^a = n^b \nabla_b \eta^a - \eta^b \nabla_b n^a, \quad (3.13)$$

and so

$$n^b \nabla_b \eta^a = B^a_b \eta^b. \quad (3.14)$$

The left hand side of Eq. (3.14) is the parallel transport of η^a along n^a , which tells us that the linear map B^a_b measures the failure of η^a to be parallel transported along the geodesics with tangent vectors n^a . Now, we can define the *expansion* θ , *shear* σ_{ab} , and *twist* ω_{ab} of the tensor B_{ab} as

$$\theta = B^{ab} h_{ab}, \quad (3.15)$$

$$\sigma_{ab} = B_{(ab)} - \frac{1}{3} \theta h_{ab}, \quad (3.16)$$

$$\omega_{ab} = B_{[ab]}, \quad (3.17)$$

which enables us to write B_{ab} as a sum of its trace, symmetric-traceless, and anti-symmetric parts via

$$B_{ab} = \frac{1}{3} \theta h_{ab} + \sigma_{ab} + \omega_{ab}. \quad (3.18)$$

We also note that since B_{ab} and h_{ab} are purely spatial, using the symmetry properties of σ_{ab} and ω_{ab} , a simple calculation shows that σ_{ab} and ω_{ab} are also purely spatial. The geometric interpretation of Eq. (3.14) is that along any geodesic in the congruence, θ measures the average expansion of the nearby geodesics, σ_{ab} measures their shear, and ω_{ab} measures their rotation. More specifically, given an initial sphere, the eigenvectors of the linear map σ^a_b give the principle axes of the ellipsoid the sphere is transported to, with the scale factors given by the eigenvalues of σ^a_b . The geodesic deviation equation can be used to provide equations of motion for all of θ , σ_{ab} , and ω_{ab} , however for our purposes we only require the equation for θ , which we derive below. To begin, first consider the following calculation:

$$\begin{aligned} n^c \nabla_c B_{ab} &= n^c \nabla_c \nabla_b n_a = n^c \nabla_b \nabla_c n_a + n^c [\nabla_c, \nabla_b] n_a \\ &= \nabla_b (n^c \nabla_c n_a) - (\nabla_b n^c) (\nabla_c n_a) + R_{cbad} n^c n^d \\ &= -B^c_b B_{ac} + R_{cbad} n^c n^d, \end{aligned} \quad (3.19)$$

where we used the definition of the Riemann tensor and the definition of B_{ab} , Eq. (3.8). Taking the trace of Eq. (3.19) gives the left-hand side:

$$\text{LHS} = n^c \nabla_c B_{ab} g^{ab} = n^c \nabla_c B_{ab} h^{ab} = n^c \nabla_c \theta = n^c \partial_c \theta = \dot{\theta}, \quad (3.20)$$

where we have used the purely spatial property of B_{ab} and the definition of θ . Taking the trace of the

right-hand side of Eq. (3.19) gives

$$\begin{aligned}
\text{RHS} &= -g_{ab}B^c{}_b B_a{}^c + g^{ab}R_{cbad}n^c n^d \\
&= -B^{ca}B_{ac} - R_{cd}n^c n^d \\
&= -\left(\frac{1}{3}\theta h^{ca} + \sigma^{ca} + \omega^{ca}\right)\left(\frac{1}{3}\theta h_{ac} + \sigma_{ac} + \omega_{ac}\right) - R_{cd}n^c n^d \\
&= -\frac{1}{9}\theta^2 h^{ca}h_{ac} - \sigma_{ca}\sigma^{ac} - \omega_{ca}\omega^{ac} - R_{cd}n^c n^d \\
&= -\frac{1}{3}\theta^2 - \sigma_{ac}\sigma^{ac} + \omega_{ac}\omega^{ac} - R_{cd}n^c n^d,
\end{aligned} \tag{3.21}$$

where, to expand the brackets in the third line, we have used $h_{ab}h^{ab} = 3$, the purely spatial properties of σ_{ab} and ω_{ab} , and the fact that $\sigma_{ab}\omega^{ab} = 0$ due to symmetry/anti-symmetry of their indices. We arrive at the following equation, called the Raychaudhuri equation

$$\dot{\theta} = -\frac{1}{3}\theta^2 - \sigma_{ab}\sigma^{ab} + \omega_{ab}\omega^{ab} - R_{ab}n^a n^b. \tag{3.22}$$

Our final aim in this subsection is rewrite Eq. (3.22) in the form as it appears as in [8]. To do this, we first note that since $K_{ab} = B_{(ab)}$, we have

$$K_{ab} = \frac{1}{3}\theta h_{ab} + \sigma_{ab}, \tag{3.23}$$

of which taking the trace gives

$$K = \theta, \tag{3.24}$$

where $K = K_{ab}h^{ab}$. Furthermore, from now on we will only consider *irrotational* congruences, that is, the twist tensor $\omega_{ab} = 0$. With these changes, the Raychaudhuri equation becomes

$$\dot{K} = -\frac{1}{3}K^2 - \sigma_{ab}\sigma^{ab} - R_{ab}n^a n^b. \tag{3.25}$$

We note that the last term in Eq. (3.25) is precisely the left-hand side of the geometric form of the SEC. As mentioned in Sec. 2, the Einstein equation is used to go between the physical and geometric formulations of the pointwise energy conditions. We note, however, that different physical and geometric formulations of the pointwise energy conditions will be found depending on whether one contains a cosmological constant term in their description of the Einstein equation. For our purposes, we include a cosmological constant Λ (later it will be assumed that Λ is positive), and so the calculation from the geometric to the physical formulation can be done as follows. Starting with

$$G_{ab} + \Lambda g_{ab} = R_{ab} - \frac{1}{2}Rg_{ab} + \Lambda g_{ab} = 8\pi T_{ab}, \tag{3.26}$$

we take the trace to obtain

$$R = 4\Lambda - 8\pi T, \tag{3.27}$$

and so

$$R_{ab}n^a n^b = -\Lambda + 8\pi\left(T_{ab} - \frac{1}{2}g_{ab}T\right)n^a n^b. \tag{3.28}$$

Substituting for $R_{ab}n^a n^b$, Eq. (3.25) becomes

$$\dot{K} = \Lambda - \frac{1}{3}K^2 - \sigma_{ab}\sigma^{ab} - 8\pi\left(T_{ab} - \frac{1}{2}g_{ab}T\right)n^a n^b, \tag{3.29}$$

which is seen in [8] as Eq. (10). The last term in Eq. (3.29) is now the left-hand side of the physical form of the SEC (multiplied by a constant), and is one of the key equations we need to prove the Cosmic No-Hair Theorem in Sec. 4.

3.3 The Gauss-Codazzi Equations

The Gauss-Codazzi equations are two equations relating the curvature of the hypersurfaces to the curvature of the full spacetime. To do this, we make use of the induced metric on the hypersurfaces h_{ab} , which induces a unique derivative operator on each hypersurface Σ_t , which we will call D_a to distinguish it from ∇_a , the covariant on the full spacetime metric g_{ab} . The operator D_a then defines a spatial curvature tensor ${}^{(3)}R_{abc}{}^d$ on each Σ_t , from which we can obtain information about the curvature of the hypersurfaces Σ_t . In what follows, we derive equations relating D_a and ${}^{(3)}R_{abc}{}^d$ to their four-dimensional counterparts ∇_a and $R_{abc}{}^d$.

To begin with, consider the linear map $h^a{}_b$. Given a vector $v^a \in T_p M$ at some point $p \in M$, we also have $p \in \Sigma_t$ for some $t \in \mathbb{R}$ due to the foliation of the spacetime, and so we can decompose v^a into components parallel and orthogonal to Σ_t via

$$v^a = v_\perp n^a + v_\parallel^a, \quad (3.30)$$

where n^a is the timelike, normalised vector orthogonal to Σ_t and $g_{ab}n^a v_\parallel^b = 0$. If $v_\perp = 0$, then $v^a = v_\parallel^a$ and v^a can be viewed as a tangent vector to Σ_t in p , i.e. $v^a \in T_p \Sigma_t$. Now, consider the following calculation:

$$h^a{}_b v^b = g^{ac}(g_{cb} + n_c n_b) v^b = (\delta^a{}_b + n^a n_b) v^b = v^a - v_\perp n^a = v_\parallel^a, \quad (3.31)$$

which tells us that $h^a{}_b$ can be thought of as a projection operator from $T_p M$ to $T_p \Sigma_t$. Hence, a spacetime vector $v^a \in T_p M$ is in $T_p \Sigma_t$ if and only if $v^a = h^a{}_b v^b$. In general, a spacetime tensor $T^{a_1 \dots a_k}{}_{b_1 \dots b_l}$ at $p \in T_p \Sigma_t \subset M$ can be viewed as a tensor over $T_p \Sigma_t$ if

$$T^{a_1 \dots a_k}{}_{b_1 \dots b_l} = h^{a_1}{}_{c_1} \dots h^{a_k}{}_{c_k} h_{b_1}{}^{d_1} \dots h_{b_l}{}^{d_l} T^{c_1 \dots c_k}{}_{d_1 \dots d_l} \quad (3.32)$$

This now motivates the use of the following result from [24]:

Proposition 1. Let (M, g_{ab}) be a spacetime and Σ_t a smooth spacelike hypersurface. Let h_{ab} be the induced spatial metric on Σ_t and let D_a be the derivative operator associated with h_{ab} . Let $T^{a_1 \dots a_k}{}_{b_1 \dots b_l}$ be a tensor field on Σ_t , then D_a is given by the formula

$$D_c T^{a_1 \dots a_k}{}_{b_1 \dots b_l} = h^{a_1}{}_{d_1} \dots h^{a_k}{}_{d_k} h_{b_1}{}^{e_1} \dots h_{b_l}{}^{e_l} h_c{}^f \nabla_f T^{d_1 \dots d_k}{}_{e_1 \dots e_l}, \quad (3.33)$$

where ∇_a is the derivative operator associated with g_{ab} .

Prop. 1 can be easily shown by verifying all the properties a derivative operator is required to possess. Most of these properties follow directly from the fact that ∇_a is a derivative operator, however to show that h_{ab} vanishes when acted on by D_a , we have the calculation

$$D_a h_{bc} = h_a{}^d h_b{}^e h_c{}^f \nabla_d (g_{ef} + n_e n_f) = 0, \quad (3.34)$$

since $\nabla_d g_{ef} = 0$, and after using the Leibniz rule to expand $\nabla_d (n_e n_f)$ we can use $h_{ab} n^a = 0$.

Continuing, if ω_a is a dual vector on Σ_t , then we can define the spatial curvature tensor ${}^{(3)}R_{abc}{}^d$ in the usual way as

$${}^{(3)}R_{abc}{}^d \omega_d = D_a D_b \omega_c - D_b D_a \omega_c. \quad (3.35)$$

Our aim is now to explicitly calculate the right-hand side of Eq. (3.35) using Eq. (3.33). Starting with $D_a D_b \omega_c$, we apply the definition of D_a twice to get

$$\begin{aligned} D_a D_b \omega_c &= D_a (h_b{}^d h_c{}^e \nabla_d \omega_e) \\ &= h_a{}^f h_b{}^g h_c{}^k \nabla_f (h_g{}^d h_k{}^e \nabla_d \omega_e) \\ &= h_a{}^f h_b{}^d h_c{}^e \nabla_f \nabla_d \omega_e + h_a{}^f h_b{}^d h_c{}^k (\nabla_d \omega_e) (\nabla_f h_k{}^e) + h_a{}^f h_b{}^d h_c{}^e (\nabla_d \omega_e) (\nabla_f h_g{}^d), \end{aligned} \quad (3.36)$$

where to obtain the third line we use the Leibniz rule for ∇_f . Considering the second term in the third line in Eq. (3.36), we have

$$\begin{aligned}
h_a^f h_c^k \nabla_f h_k^e &= h_a^f h_c^k \nabla_f (\delta_k^e + n_k n^e) \\
&= (h_a^f \nabla_f n^e) h_c^k n_k + (h_a^f \nabla_f n_k) h_c^k n^e \\
&= K_{ak} h_c^k n^e \\
&= K_{ak} (\delta_c^k + n_c n^k) n^e \\
&= K_{ac} n^e,
\end{aligned} \tag{3.37}$$

where in the second line we have used $h_{ab} n^b = 0$ and the definition of K_{ab} , and in the fifth line we have used $K_{ab} n^b = 0$. A similar calculation in the third term in the third line of Eq. (3.36) gives

$$h_a^f h_b^g \nabla_f h_g^d = K_{ab} n^d, \tag{3.38}$$

which we observe will vanish when anti-symmetrised over a and b . Now, in the second term in the third line in Eq. (3.36) we now have

$$h_b^d n^e \nabla_d \omega_e = h_b^d \nabla_d (n^e \omega_e) - h_b^d \omega_e \nabla_d n^e = -K_b^e \omega_e, \tag{3.39}$$

where we have used $n^e \omega_e = 0$ on Σ_t and the definition of K_{ab} . We arrive at

$$D_a D_b \omega_c = h_a^f h_b^g h_c^k \nabla_f \nabla_g (h_k^d \omega_d) - K_{ac} K_b^d \omega_d \tag{3.40}$$

where we have replaced ω_k with $h_k^d \omega_d$ which ensure ∇_a acts on a spacetime tensor. We can now anti-symmetrise over a and b , giving

$$\begin{aligned}
{}^{(3)}R_{abc}{}^d \omega_d &= h_a^f h_b^g h_c^k [\nabla_f, \nabla_g] h_k^d - K_{ac} K_b^d \omega_d + K_{bc} K_a^d \omega_d \\
&= h_a^f h_b^g h_c^k R_{fgk}{}^j h_j^d \omega_d - K_{ac} K_b^d \omega_d + K_{bc} K_a^d \omega_d.
\end{aligned} \tag{3.41}$$

Cancelling the arbitrary ω_d , we finally arrive at

$${}^{(3)}R_{abc}{}^d = h_a^f h_b^g h_c^k h_j^d R_{fgk}{}^j - K_{ac} K_b^d + K_{bc} K_a^d. \tag{3.42}$$

Eq. (3.42) is one of the two *Gauss-Codazzi equations* and relates the spatial curvature of the hypersurfaces to the Riemann curvature of the spacetime. For our purposes we will only need Eq. (3.42), however for completeness, the second Gauss-Codazzi equation, relating the spatial curvature of the hypersurfaces to the Ricci curvature of the spacetime, is given by

$$D_a K_b^a - D_b K_a^a = R_{cd} n^d h_b^c. \tag{3.43}$$

The final step in this subsection is to use Eq. (3.42) to obtain an equation relating K , ${}^{(3)}R$ and a term resembling a pointwise energy condition, which we will see to be $T_{ab} n^a n^b$. To begin this process, consider the calculation

$$\begin{aligned}
h^{ac} h^{bd} R_{abcd} &= (g^{ac} + n^a n^b)(g^{bd} + n^b n^d) R_{abcd} \\
&= R + 2R_{ab} n^a n^b \\
&= 2(R_{ab} - \frac{1}{2} R g_{ab}) n^a n^b \\
&= 2G_{ab} n^a n^b,
\end{aligned} \tag{3.44}$$

where the last term after expanding the brackets on the first line vanishes due to the antisymmetry of a and b in R_{abcd} and the symmetry of a and b in $n^a n^b$. Next, we contract b and d in Eq. (3.42) and

multiply by g^{ac} to get

$$\begin{aligned}
{}^{(3)}R &= g^{ac}h_a^f h_b^g h_c^k h_j^b R_{fgk}^j - K^2 + K_{ab}K^{ab} \\
&= h^{cf}h_c^k h_j^g R_{fgk}^j - K^2 + K_{ab}K^{ab} \\
&= (g^{cf} + n^c n^f)h_c^k h_j^g R_{fgk}^j - K^2 + K_{ab}K^{ab} \\
&= h^{fk}h^{jg}R_{fgkj} - K^2 + K_{ab}K^{ab},
\end{aligned} \tag{3.45}$$

where ${}^{(3)}R$ is defined as ${}^{(3)}R = {}^{(3)}R_{ab}g^{ab}$ and ${}^{(3)}R_{ab} = {}^{(3)}R_{acb}{}^c$. Equating the terms $h^{ac}h^{bd}R_{abcd}$ in Eq. (3.44) and Eq. (3.45), we obtain

$$G_{ab}n^a n^b = \frac{1}{2}({}^{(3)}R + K^2 - K_{ab}K^{ab}). \tag{3.46}$$

Next, using Eq. (3.23), we substitute for K_{ab} in Eq. (3.46) to obtain

$$\begin{aligned}
G_{ab}n^a n^b &= \frac{1}{2}\left[{}^{(3)}R + K^2 - \left(\frac{1}{3}Kh_{ab} + \sigma_{ab}\right)\left(\frac{1}{3}Kh^{ab} + \sigma^{ab}\right)\right] \\
&= \frac{1}{2}({}^{(3)}R + \frac{2}{3}K^2 - \sigma_{ab}\sigma^{ab}).
\end{aligned} \tag{3.47}$$

We note that $G_{ab}n^a n^b$ is precisely the geometric form of the WEC, and so to obtain an expression involving the physical form of the WEC, we follow the same steps as in Subsec. (3.2). Substituting Eq. (3.47) into Eq. (3.26) multiplied by $n^a n^b$, we obtain

$$K^2 = 3\Lambda + \frac{3}{2}\sigma_{ab}\sigma^{ab} - \frac{3}{2}{}^{(3)}R + 24\pi T_{ab}n^a n^b, \tag{3.48}$$

which is seen in [8] as Eq. (9). The last term in Eq. (3.48) is now the physical form of the WEC (multiplied by a constant) and is another one of the key equations we need to prove the Cosmic No-Hair Theorem in Sec. 4.

3.4 The Bianchi Classification

With Eqs. (3.29) and (3.48) derived, the next step towards the Cosmic No-Hair Theorem is to consider the types of universes that can occur when we demand spatial homogeneity in our 1+3 dimensional spacetime. This can be done by finding all (real) three-dimensional Lie algebras and was first accomplished in 1897 by Bianchi in [29] (translation). This list, now called the ‘‘Bianchi Classification’’ contains nine Lie algebras, which are in one-to-one correspondence with three-dimensional simply connected Lie groups. In the context of cosmology, these Lie algebras classify all the types of spatially homogeneous 1+3 dimensional spacetimes. Often called the ‘‘Bianchi universes’’, they are the Lie algebras of the spatial symmetry groups of three-dimensional Riemannian manifolds, which include our spacelike hypersurfaces. In this subsection, we shall derive the classification using the method from [24], which itself is a modified derivation from [30].

To begin, we note that finding all three-dimensional Lie algebras is equivalent to finding all possible tensors C_{ab}^c , where we demand antisymmetry in the lower indices, $C_{ab}^c = -C_{ba}^c$, and that they satisfy the Jacobi identity

$$C_{d[a}^e C_{bc]}^d = 0. \tag{3.49}$$

These tensors C_{ab}^c can then be identified as the structure constants of a three-dimensional Lie algebra on a vector space V , since they will then uniquely specify a Lie algebra. Now, let ϵ_{abc} be a fixed totally antisymmetric (0,3) tensor and ϵ^{abc} be the unique totally antisymmetric (3,0) tensor such that $\epsilon_{abc}\epsilon^{abc} = 3! = 6$. Now given C_{ab}^c , define the tensors A_a and M^{ab} by

$$A_a = C_{ba}^b, \tag{3.50}$$

$$M^{ab} = \frac{1}{2}\epsilon^{acd}(C^b_{cd} - \delta^b_c A_d). \quad (3.51)$$

Contracting Eq. (3.51) with ϵ_{aef} and using the formula

$$\epsilon_{aef}\epsilon^{acd} = 2\delta^{[c}_e\delta^d]_f, \quad (3.52)$$

we obtain

$$C^b_{ef} = M^{ab}\epsilon_{aef} + \delta^b_{[e}A_{f]}. \quad (3.53)$$

Next, contracting Eq. (3.53) over b, e we find that

$$\epsilon_{abf}M^{ab} = 0 \iff M^{[ab]} = 0, \quad (3.54)$$

i.e. M^{ab} is symmetric. Changing indices in Eq. (3.53) we can now write C^c_{ab} as

$$C^c_{ab} = M^{cd}\epsilon_{dab} + \delta^c_{[a}A_{b]}, \quad (3.55)$$

where $M^{ab} = M^{ba}$, and Eq. (3.53) is manifestly antisymmetric in its lower indices. Imposing the Jacobi identity Eq. (3.49) on Eq. (3.55) gives the relation

$$M^{ab}A_b = 0. \quad (3.56)$$

Eq. (3.56) tells us that all three-dimensional Lie-algebras are determined by the dual vector A_a and the symmetric $(2,0)$ tensor M^{ab} such that $M^{ab}A_b=0$. We can classify these Lie algebras (up to isomorphism) as follows. The first case, which we call “class A” is the case in which $A_a = 0$ and causes Eq. (3.56) to be immediately satisfied. Hence, the Lie algebras in class A are determined by the rank and signature of M^{ab} (up to a sign), for which there are six possible cases. The second case, which we call “class B” is the case in which $A_a \neq 0$. In this case, the Lie algebras are still determined by the rank and signature of M^{ab} (up to a sign), however the rank of M^{ab} cannot be greater than two, since if M^{ab} has full rank, Eq. (3.56) cannot be satisfied with $A_a \neq 0$. There exist four possibilities for M^{ab} in this case. Furthermore, in the case where the rank of M^{ab} is two, we can define a scalar α by

$$A_e A_f = \alpha M^{ac} M^{bd} \epsilon_{cde} \epsilon_{abf}, \quad (3.57)$$

which is well-defined since both tensors on either side of Eq. (3.57) are non-vanishing. Hence, for these two cases we have a continuum of Lie algebras determined by the scalar α , and we also note that the two cases where M^{ab} has rank two in class A can also be considered to be parts of these continua (in their respective signatures) if we take $\alpha = 0$. Finally, we also distinguish the case $\alpha = -1$ when M^{ab} has rank two and signature $+-$ in class B. Table 1 lists all the possible Lie algebras according to this prescription and can also be found in a slightly alternate form in [30], while examples of specific choices for the components of A_a and M^{ab} can be found in [31].

For the remainder of the subsection, we will use this classification to gain insight into the spatial geometry of the hypersurfaces. In particular, we will show that the spatial curvature ${}^{(3)}R$ is non-positive in all Bianchi universes except type IX. To do this, we first note that by using Eq. (7.2.41) in [24] and Eq. (3.46), we can write ${}^{(3)}R$ in terms of the structure-constants C^c_{ab} as

$${}^{(3)}R = -C^a_{ab}C^c_c{}^b + \frac{1}{2}C^a_{bc}C^c_{ab} - \frac{1}{4}C_{abc}C^{abc}, \quad (3.58)$$

where indices are lowered and raised using h_{ab} and h^{ab} . Next, we substitute Eq. (3.55) into Eq. (3.58) to obtain

$${}^{(3)}R = -\frac{3}{2}A_b A^b - h^{-1}\left(M_{ab}M^{ab} - \frac{1}{2}M^2\right), \quad (3.59)$$

where $h^{-1} = \det(h^{ab})$. This calculation is relatively involved and can be found in Appendix A. From Eq. (3.59), we can see that if ${}^{(3)}R > 0$, then $M_{ab}M^{ab} < \frac{1}{2}M^2$, and so using [32], we have that M^{ab} is positive definite (or negative definite). From Table 1, we see that this corresponds exactly to a type IX Bianchi universe. Taking the contrapositive, we have the result that in all Bianchi universes except type IX,

$${}^{(3)}R \leq 0. \quad (3.60)$$

Class	Rank of M^{ab}	Signature of M^{ab}	Bianchi Type
A	0		I
	1	+	II
	2	+ +	VII ($\alpha = 0$)
	2	+ -	VI ($\alpha = 0$)
	3	+ + +	IX
	3	+ + -	VIII
B	0		V
	1	+	IV
	2	+ +	VII ($\alpha \neq 0$)
	2	+ -	VI ($\alpha \neq 0$), (III if $\alpha = -1$)

Table 1: The nine types of Bianchi universes, where we have split the type VI and type VII into the cases $\alpha = 0$ and $\alpha \neq 0$ and have also distinguished the case $\alpha = -1$ as type III. For every number of signs less than three that appear in the signature of M^{ab} column, we mean that M^{ab} has an eigenvector with eigenvalue 0.

4 The Cosmic No-Hair Theorem

Now we have the main tools to prove the Cosmic No-Hair Theorem, which we will state first and prove in the remainder of this section.

Theorem 1. *Cosmic No-Hair Theorem.* Let (M, g_{ab}) be a homogeneous, initially expanding spacetime satisfying Einstein's equation with positive cosmological constant Λ . Assuming the stress-energy tensor satisfies the dominant and strong energy conditions, then for all Bianchi universes except type IX, the spacetime locally approaches de Sitter with time scale $(3/\Lambda)^{1/2}$.

To begin, considering Eq. (3.48), using $\sigma_{ab}\sigma^{ab} \geq 0$ (since its indices are spatial), Eq. (3.60) and the WEC (recall that the DEC implies the WEC), we obtain the inequality

$$K^2 \geq 3\Lambda. \quad (4.1)$$

Since we assume that at an initial time $\tau_0 = 0$ we have $K_0 > 0$, we must have $K > 0$ at all future times i.e.

$$K \geq (3\Lambda)^{1/2}. \quad (4.2)$$

In other words, if the universe is initially expanding, then it will remain expanding for all future times.

Next, considering Eq. (3.29), again using $\sigma_{ab}\sigma^{ab} \geq 0$, and the SEC, we obtain

$$\dot{K} \leq \Lambda - \frac{1}{3}K^2, \quad (4.3)$$

which we can then combine with Eq. (4.1) to obtain

$$\dot{K} \leq \Lambda - \frac{1}{3}K^2 \leq 0. \quad (4.4)$$

Now, writing Eq. (4.3) as

$$\frac{1}{K^2 - 3\Lambda} \dot{K} \leq -\frac{1}{3}, \quad (4.5)$$

which we can integrate from 0 to τ along a future complete geodesic in our timelike congruence to get

$$\coth^{-1}\left(\frac{K}{(3\Lambda)^{1/2}}\right) \geq \frac{\tau}{\alpha} + \coth^{-1}\left(\frac{K_0}{(3\Lambda)^{1/2}}\right) \geq \frac{\tau}{\alpha}, \quad (4.6)$$

where $\alpha = (3/\Lambda)^{1/2}$ and the second inequality holds since $K_0 > 0$. Writing Eq. (4.6) in terms of K gives

$$K \leq \frac{(3\Lambda)^{1/2}}{\tanh(\tau/\alpha)}, \quad (4.7)$$

where the direction of the inequality reverses since $\coth$ is decreasing. Combining Eq. (4.2) and Eq. (4.7) gives the inequality

$$(3\Lambda)^{1/2} \leq K \leq \frac{(3\Lambda)^{1/2}}{\tanh(\tau/\alpha)}, \quad (4.8)$$

and so we see that as $\tau \rightarrow \infty$, K is squeezed exponentially to the value $(3\Lambda)^{1/2}$ with time scale α . Similarly, from Eq. (3.48) we get

$$\sigma_{ab}\sigma^{ab} \leq \frac{2}{3}(K^2 - 3\Lambda) \leq \frac{2\Lambda}{\sinh^2(\tau/\alpha)}, \quad (4.9)$$

where the second inequality comes from substituting the upper bound for K from Eq. (4.7). Hence, since the shear has spatial indices, we have that $\sigma_{ab} \rightarrow 0$ with time scale α . Also from Eq. (3.48), we can obtain

$$T_{ab}n^an^b \leq \frac{1}{24\pi}(K^2 - 3\Lambda) \leq \frac{\Lambda}{8\pi} \frac{1}{\sinh^2(\tau/\alpha)}, \quad (4.10)$$

which implies that $T_{ab}n^an^b \rightarrow 0$ with time scale α . Since we interpret $T_{ab}n^an^b$ as the energy density ρ measured by an observer travelling on the timelike geodesic, assuming the DEC, we have that the principal pressures P_i approach zero since $\rho \geq |P_i|$. Recalling we can write T_{ab} in the form of Eq. (2.1), the result is that all the components of T_{ab} approach zero. Next, from Eq. (3.23) we have that $K_{ab} \rightarrow h_{ab}/\alpha$ and so from Eq. (3.6), at late times h_{ab} satisfies the ODE

$$\dot{h}_{ab} = \frac{2}{\alpha}h_{ab}, \quad (4.11)$$

and so at late times the spatial metric is given by

$$h_{ab}(\tau) = e^{2\tau/\alpha}h_{ab}(0). \quad (4.12)$$

From Eq. (4.12), we see that the induced metric at late times depends only on τ , not the position inside any given hypersurface Σ_τ . In other words, each Σ_τ is “isotropic”, meaning that space looks the same in any direction from any point of observation and also implies that the hypersurfaces become locally flat and expand at the rate $K = (3\Lambda)^{1/2}$. With the addition that at late times we have a vanishing stress tensor, and so the universe appear to be matter free, we conclude that the universe approaches de Sitter spacetime with time scale $\alpha = (3/\Lambda)^{1/2}$ and hence we have proven theorem 1.

5 Averaged energy conditions

An *averaged energy condition* is a condition on the stress-energy tensor T_{ab} of a theory averaged over some spacetime region. In principle, we can pick any spacetime region to average over, but it is most common to choose causal curves (spacetime curves in which all tangent vectors are timelike or null). Like the pointwise energy conditions, the averaged energy condition can be given in a physical, geometric and effective formulation. Similarly as before, assuming Einstein’s equation holds, the physical and geometric formulations are equivalent and furthermore, assuming the stress-energy tensor is of the form of Eq. (2.1), i.e. it is type I, the effective formulation is also equivalent to the two others. Below, we state the most common averaged energy conditions in the physical and geometric formulation. An effective formulation requires slightly more technical detail and will be omitted here, and one can refer to [11] for its full description.

Let τ and λ be affine parameters, where τ will parametrise timelike curves and so, situation depending, can be taken to be the proper time. λ will parametrise null curves so it will remain as an affine parameter. In n spacetime dimension, the most common averaged energy conditions are:

Averaged weak energy conditon (AWEC)

physical for all inextendible timelike curves γ with tangent vector ξ^a ,

$$\int_{\gamma} d\tau T_{ab} \xi^a \xi^b \geq 0 \quad (5.1)$$

geometric for all inextendible timelike curves γ with tangent vector ξ^a ,

$$\int_{\gamma} d\tau G_{ab} \xi^a \xi^b \geq 0 \quad (5.2)$$

Averaged strong energy conditon (ASEC)

physical for all inextendible timelike curves γ with tangent vector ξ^a ,

$$\int_{\gamma} d\tau \left(T_{ab} - \frac{1}{n-2} T g_{ab} \right) \xi^a \xi^b \geq 0 \quad (5.3)$$

geometric for all inextendible timelike curves γ with tangent vector ξ^a ,

$$\int_{\gamma} d\tau R_{ab} \xi^a \xi^b \geq 0 \quad (5.4)$$

Averaged null energy conditon (ANEC)

physical for all inextendible null curves γ with tangent vector k^a ,

$$\int_{\gamma} d\lambda T_{ab} k^a k^b \geq 0 \quad (5.5)$$

geometric for all inextendible null curves γ with tangent vector k^a ,

$$\int_{\gamma} d\lambda R_{ab} k^a k^b \geq 0 \quad (5.6)$$

All the conditions above are taken to hold when the integral is absolutely convergent, and the integrands are understood to be evaluated along the causal curve γ . One final condition worth mentioning is the *averaged achronal null energy condition* (AANEC), in which the conditions of the ANEC hold with the additional requirement that γ is *achronal* (meaning no two points on γ are separated by a timelike curve). As previously mentioned, the AANEC is widely believed to be the averaged energy condition that holds the most generally. The AANEC holds under many quantum fields, and where it fails, it does so for reasons that the generally seen as unphysical. These include situations where equations of motion, like Einstein's equation, are broken, or where physical quantities take on trans-Planckian values.

There does not exist any meaningful interpretations for the averaged energy conditions given above, and one can refer to [11] for a detailed discussion of this fact. We can however provide some motivation for extending the pointwise energy conditions to averaged ones. Noticing that the conditions above are merely the corresponding pointwise conditions integrated over a causal geodesic, we interpret this as allowing small local violations of the pointwise conditions, but over a large spacetime volume, the average must be non-negative. We can further interpret that violations of the pointwise energy conditions are caused by quantum effects which are small in magnitude and duration, and again these violations disappear when we average over a whole causal geodesic.

The averaged energy conditions above are weaker than their pointwise counterparts and so, in general, they are less frequently violated. As mentioned, the non-minimally coupled scalar field violates all the main pointwise energy conditions, while in Ricci-flat spacetimes, the ANEC and AWEC are satisfied for certain values of the coupling constant. In the following section, we prove these results and also discuss results pertaining to the ASEC.

6 The non-minimally coupled scalar field

The non-minimally coupled scalar field connects a scalar field ϕ directly to the curvature of a spacetime R via a coupling constant ξ . This provides a more complex interaction between the field and the curvature of spacetime than the normal minimally coupled scalar field. From [33] (adjusting for our sign convention), we take the Lagrangian for the non-minimally coupled scalar field ϕ to be

$$\mathcal{L}[\phi] = \frac{1}{2}[(\nabla\phi)^2 + (m^2 + \xi R)\phi^2], \quad (6.1)$$

where $m \geq 0$ is the mass and ξ is the coupling constant to the scalar curvature R . From the Lagrangian, we can obtain the Klein-Gordon equation of motion

$$(-\square + m^2 + \xi R)\phi = 0. \quad (6.2)$$

The stress tensor for this theory is obtained as usual by varying the action with respect to the inverse metric:

$$T_{ab} = (\nabla_a\phi)(\nabla_b\phi) - \frac{1}{2}g_{ab}((\nabla\phi)^2 + m^2\phi^2) + \xi(g_{ab}\square - \nabla_a\nabla_b + G_{ab})\phi^2, \quad (6.3)$$

where $\square = \nabla_a\nabla^a$ denotes the d'Alembert operator. Now with the stress tensor calculated, we show that the SEC is easily violated in this theory as follows. Taking the trace of Eq. (6.3) we obtain

$$T = \left(1 - \frac{n}{2}\right)(\nabla\phi)^2 - \frac{n}{2}m^2\phi^2 + \xi\left((n-1)\square + \left(1 - \frac{n}{2}\right)R\right)\phi^2, \quad (6.4)$$

and we also define the *effective energy density* (EED) ρ as

$$\rho = T_{ab}u^au^b + \frac{T}{n-2}, \quad (6.5)$$

where u^a is a unit timelike velocity field. Here, we use u^a instead of n^a for a timelike vector, which is done to highlight how u^a is not necessarily the normal vector of a foliation of the spacetime. We note that ρ is just the left hand side of the SEC with the addition that u^a is normalised. Substituting Eq. (6.3) and Eq. (6.4) into Eq. (6.5) gives the EED

$$\begin{aligned} \rho = & (1 - 2\xi)(\nabla_u\phi)^2 - \frac{1 - 2\xi}{n-2}m^2\phi^2 + \frac{2\xi}{n-2}(\nabla\phi)^2 \\ & - 2\xi\phi(\nabla_u^2\phi) + \xi R_{ab}u^au^b\phi^2 + \frac{2\xi^2}{n-2}R\phi^2 - \frac{2\xi}{n-2}(\phi P\phi), \end{aligned} \quad (6.6)$$

where $\nabla_u := u^a\nabla_a$, $\nabla_u^2 := u^au^b\nabla_a\nabla_b$, and $P := -\square + m^2 + \xi R$, and so the final term vanishes if we take ϕ “on-shell”, i.e. satisfying Eq. (6.2). Now, if we take $\xi = 0$, the EED becomes

$$\rho = (\nabla_u\phi)^2 - \frac{1}{n-2}m^2\phi^2, \quad (6.7)$$

and so the SEC is violated if at any point we have $m^2\phi^2 > (n-2)(\nabla_u\phi)^2$, which can occur if either m or ϕ become arbitrarily large or if the derivatives of ϕ along u^a vanish.

6.1 A strong worldline energy inequality

Continuing from our expression for the EED, Eq. (6.6), our aim now is to derive some sort of energy condition along the worldline of a timelike geodesic, and we follow the arguments from [33] (in our conventions). This is not exactly the ASEC, but the techniques used do give insight into how the EED behaves when average over timelike curves. We will let γ be a unit timelike geodesic with tangent vector $\dot{\gamma}^a = u^a$ parametrised by proper time τ , and let $f \in C_0^2(\mathbb{R})$ be a real-valued compactly supported function. We consider the expression

$$\int_{\gamma} d\tau \rho f^2(\tau), \quad (6.8)$$

which we want to provide a lower bound for. Here, f is sometimes called a “smearing function” whose purpose includes a way of controlling the average of ρ over an entire geodesic, and also enables the use of certain scaling arguments, as we will see below. Using the expression for the EED ρ Eq. (6.6) above, “on-shell” we have that

$$\begin{aligned} \int_{\gamma} d\tau \rho f^2(\tau) &= \int_{\gamma} d\tau \left((1 - 2\xi)(\nabla_u \phi)^2 - \frac{1 - 2\xi}{n - 2} m^2 \phi^2 + \frac{2\xi}{n - 2} (\nabla \phi)^2 \right. \\ &\quad \left. - 2\xi \phi (\nabla_u^2 \phi) + \xi R_{ab} u^a u^b \phi^2 + \frac{2\xi^2}{n - 2} R \phi^2 \right) f^2(\tau). \end{aligned} \quad (6.9)$$

We now consider Eq. (9) in [14], which is an identity used in quantum mechanics and is derived in [14] via vector calculus. We integrate this equality to obtain

$$-2\xi \int_{\gamma} d\tau f^2(\tau) \phi (\nabla_u^2 \phi) = 2\xi \int_{\gamma} [\nabla_u (f(\tau) \phi)]^2 - 2\xi \int_{\gamma} d\tau \phi^2 (f'(\tau))^2, \quad (6.10)$$

which we have used to write the term containing $\phi (\nabla_u^2 \phi)$ as a difference of two non-negative terms. We also define $h^{ab} = g^{ab} + u^a u^b$ (as we have used previously), and note that now two of our terms in Eq. (6.9) can be written as a sum of two non-negative terms when $\xi \in [0, 2\xi_c]$ via

$$(1 - 2\xi)(\nabla_u \phi)^2 + \frac{2\xi}{n - 2} (\nabla \phi)^2 = \left(1 - \frac{\xi}{2\xi_c}\right) (\nabla_u \phi)^2 + \frac{2\xi}{n - 2} h^{ab} \nabla_a \phi \nabla_b \phi, \quad (6.11)$$

where ξ_c is called the *conformal coupling constant*, defined as

$$\xi_c = \frac{n - 2}{4(n - 1)}. \quad (6.12)$$

Recall that h_{ab} is a Riemannian metric, so the final term in Eq. (6.11) is non-negative. Substituting Eq. (6.10) and Eq. (6.11) into Eq. (6.9), we obtain

$$\begin{aligned} \int_{\gamma} d\tau \rho f^2(\tau) &= \int_{\gamma} d\tau \left(\left(1 - \frac{\xi}{2\xi_c}\right) (\nabla_u \phi)^2 + \frac{2\xi}{n - 2} h^{ab} \nabla_a \phi \nabla_b \phi - \frac{1 - 2\xi}{n - 2} m^2 \phi^2 \right. \\ &\quad \left. + \xi R_{ab} u^a u^b \phi^2 + \frac{2\xi^2}{n - 2} R \phi^2 \right) f^2(\tau) + 2\xi [\nabla_u (f(\tau) \phi)]^2 - 2\xi \phi^2 (f'(\tau))^2, \end{aligned} \quad (6.13)$$

which contains three non-negative terms and so we obtain the following inequality, seen as Theorem 1 in [33].

Theorem 2. Let γ be a timelike geodesic with unit tangent vector $\dot{\gamma}^a = u^a$ parametrised by proper time τ on a spacetime (M, g_{ab}) of dimension $n > 2$. Let T_{ab} be the stress tensor of a non-minimally coupled scalar field with coupling constant $\xi \in [0, 2\xi_c]$ and EED ρ . Then “on-shell”,

$$\begin{aligned} \int_{\gamma} d\tau \rho f^2(\tau) &\geq - \int_{\gamma} d\tau \left(\frac{1 - 2\xi}{n - 2} m^2 f^2(\tau) + 2\xi (f'(\tau))^2 - \xi R_{ab} U^a U^b f^2(\tau) \right. \\ &\quad \left. - \frac{2\xi^2}{n - 2} R f^2(\tau) \right) \phi^2. \end{aligned} \quad (6.14)$$

While Theorem 2 is not exactly the ASEC, it can be thought of as a sort of averaged strong energy condition for the non-minimally coupled scalar field. One advantage of Theorem 2 is that there are no terms containing derivatives of ϕ , which we will explore some consequences of below. We also note that, as mentioned, Eq. (6.13) contains three non-negative terms, so depending on the situation we have slightly stronger versions of Eq. (6.14). In particular, Theorem 2 gives that in Ricci flat spacetimes (where $R_{ab} = 0$), we have

$$\int_{\gamma} d\tau \rho f^2(\tau) \geq - \int_{\gamma} d\tau \left(\frac{1-2\xi}{n-2} m^2 f^2(\tau) + 2\xi (f'(\tau))^2 \right) \phi^2, \quad (6.15)$$

while in minimally coupled spacetimes (where $\xi = 0$), we have

$$\int_{\gamma} d\tau \rho f^2(\tau) \geq - \frac{1}{n-2} m^2 \int_{\gamma} d\tau f^2(\tau) \phi^2, \quad (6.16)$$

regardless of curvature. Let us explore the consequences of the flat spacetime bound Eq. (6.15). Let $\phi_{\max}$ be the maximum value of ϕ over the geodesic γ (which we assume to exist), i.e.

$$\phi_{\max} = \sup_{\gamma} |\phi|, \quad (6.17)$$

then, Eq. (6.15) becomes

$$\int_{\gamma} d\tau \rho f^2(\tau) \geq -\phi_{\max}^2 \int_{\gamma} d\tau \frac{1-2\xi}{n-2} m^2 f^2(\tau) + 2\xi (f'(\tau))^2. \quad (6.18)$$

Next, we investigate the effects of scaling the smearing function f in Eq. (6.18). We will assume that f has unit L^2 norm and define the *rescaled smearing function* $f_{\lambda}(\tau)$ by

$$f_{\lambda}(\tau) = \frac{f(\tau/\lambda)}{\sqrt{\lambda}}, \quad \lambda > 0, \quad (6.19)$$

which ensures that $f_{\lambda}(\tau)$ also has unit L^2 norm. Now, Eq. (6.18) becomes

$$\int_{\gamma} d\tau \rho f_{\lambda}^2(\tau) \geq -\phi_{\max}^2 \int_{\gamma} d\tau \frac{1-2\xi}{\lambda} \left(\frac{1-2\xi}{n-2} m^2 f^2(\tau/\lambda) + \frac{2\xi}{\lambda^2} (f'(\tau/\lambda))^2 \right), \quad (6.20)$$

which after the substitution $\tau \rightarrow \lambda\tau$ becomes

$$\int_{\gamma} d\tau \rho f_{\lambda}^2(\tau) \geq -\phi_{\max}^2 \int_{\gamma} d\tau \frac{1-2\xi}{n-2} m^2 f^2(\tau) + \frac{2\xi}{\lambda^2} (f'(\tau))^2. \quad (6.21)$$

If we take $\lambda \rightarrow 0$, $f_{\lambda}^2(\tau)$ approaches a delta function, so we can think of the left-hand side of Eq. (6.21) as just $\rho(\gamma(0))$. Since the right-hand side of Eq. (6.21) approaches negative infinity, this confirms that pointwise, the SEC can be violated arbitrarily badly. Taking $\lambda \rightarrow \infty$, we obtain

$$\liminf_{\lambda \rightarrow \infty} \int_{\gamma} d\tau \rho f_{\lambda}^2(\tau) \geq - \frac{1-2\xi}{n-2} \phi_{\max}^2 m^2, \quad (6.22)$$

which can be thought of as a lower bound for the long-term average of ρ and shows that this long-term average can be negative. Another approach seen in [33] is to estimate the supremum of ρ over an interval I of length τ_0 . Denoting $\|f\|^2$ as the L^2 norm of f over the interval I , we have

$$\sup_{\gamma(I)} \rho \|f\|^2 \geq \int_{\gamma(I)} d\tau \rho f^2(\tau) \geq - \left(\frac{1-2\xi}{n-2} m^2 \|f\|^2 + 2\xi \|f'\|^2 \right) \sup_{\gamma(I)} \phi^2, \quad (6.23)$$

and so

$$\sup_{\gamma(I)} \rho \geq - \left(\frac{1-2\xi}{n-2} m^2 + \frac{2\xi \pi^2}{\tau_0^2} \right) \sup_{\gamma(I)} \phi^2, \quad (6.24)$$

where we use that

$$\inf_f \frac{\|f'\|^2}{\|f\|^2} = \frac{\pi^2}{\tau_0^2}, \quad (6.25)$$

where the infimum ranges over all smooth f with compact support in an interval I of length τ_0 .

6.2 A weak worldline inequality

For our second example, also an argument from [33] (with our conventions), we will see that for the non-minimally coupled scalar field, it is possible under certain conditions to derive the ANEC and AWEC. To begin, we will consider only T_{ab} , which from Eq. (6.2) and Eq. (6.3) we can write ‘on-shell’ as

$$T_{ab} = (1 - 2\xi)(\nabla_a \phi)(\nabla_b \phi) - \frac{1}{2}(1 - 4\xi)g_{ab}(m^2 \phi^2 + (\nabla \phi)^2) \quad (6.26)$$

$$- \xi(2\phi \nabla_a \nabla_b \phi - R_{ab} \phi^2) - \frac{1}{2}(1 - 4\xi)g_{ab} \xi R \phi^2. \quad (6.27)$$

We will consider the expression

$$\int_{\gamma} d\lambda T_{ab} u^a u^b, \quad (6.28)$$

where γ is a causal geodesic (a geodesic with space-like or null tangent vector) with affine parameter λ and we will take $u \in \mathcal{C}_0^2(TM)$, a tangent vector field to γ . This means we can find an $f \in \mathcal{C}_0^2(\mathbb{R})$ such that $u = f\dot{\gamma}$. A calculation then shows that $u^a u^b \nabla_a \nabla_b \phi = f^2 \partial_{\lambda}^2 \phi$. Substituting Eq. (6.27) into Eq. (6.28) then gives

$$\begin{aligned} \int_{\gamma} d\lambda T_{ab} u^a u^b &= \int_{\gamma} d\lambda \left(\frac{1}{2} f^2 (\partial_{\lambda} \phi)^2 + \frac{1}{2} (1 - 4\xi) (h^{ab} (\nabla_a \phi)(\nabla_b \phi) - u^2 m^2 \phi^2) \right. \\ &\quad \left. - 2\xi f^2 \phi \partial_{\lambda}^2 \phi + \xi f^2 \phi^2 R_{ab} \dot{\gamma}^a \dot{\gamma}^b - \frac{1}{2} (1 - 4\xi) \xi f^2 \phi^2 R \dot{\gamma}^2 \right), \end{aligned} \quad (6.29)$$

where we introduce $h^{ab} = u^a u^b - u^2 g^{ab}$, which we have seen before as a Riemannian metric on each $T\Sigma$ in the foliation of M and so is a positive semi-definite metric on TM . Note that if u was time-like and normalised to $u^2 = -1$ then it is the same as what has been used previously. Now we see that when $\xi \leq 1/4$ the first two terms in Eq. (6.29) are non-negative. Next, focusing on the third term, whilst it can take any sign, it can also be written as the difference of positive terms. From [14] we have the equality

$$2f^2 \phi \partial_{\lambda}^2 \phi + \partial_{\lambda} \{ \phi \partial_{\lambda} (f^2 \phi) - f^2 \phi \partial_{\lambda} \phi \} = 2\partial_{\lambda} \{ f \phi \partial_{\lambda} (f \phi) \} - 2[\partial_{\lambda} (f \phi)]^2 + 2\phi^2 (\partial_{\lambda} f)^2, \quad (6.30)$$

which after integrating and multiplying by $-\xi$ becomes

$$-2\xi \int_{\gamma} d\lambda f^2 \phi \partial_{\lambda}^2 \phi = 2\xi \int_{\gamma} d\lambda [\partial_{\lambda} (f \phi)]^2 - 2\xi \int_{\gamma} d\lambda \phi^2 (\partial_{\lambda} f)^2. \quad (6.31)$$

This is a difference of non-negative terms if ξ is non-negative. Substituting this into Eq. (6.29) gives the following theorem, seen in [14] as Theorem II.1.

Theorem 3. Let γ be a causal geodesic with affine parameter λ in spacetime (M, g) . Let T_{ab} be the stress tensor to the non-minimally coupled scalar field with coupling constant $\xi \in [0, 1/4]$. Then for any function $f \in \mathcal{C}_0^2(\mathbb{R})$ we have ‘on-shell’ that

$$\int_{\gamma} d\lambda T_{ab} \dot{\gamma}^a \dot{\gamma}^b f^2 \geq -2\xi \int_{\gamma} d\lambda \left((\partial_{\lambda} f)^2 - \frac{1}{2} R_{ab} \dot{\gamma}^a \dot{\gamma}^b f^2 + \left(\frac{1}{4} - \xi \right) R \dot{\gamma}^2 f^2 \right) \phi^2. \quad (6.32)$$

Note that Theorem 3 also includes the case when $\xi = \xi_c$, the conformal coupling constant, defined above. To explore this further we can consider the case where (M, g_{ab}) satisfies the vacuum Einstein equations with vanishing cosmological constant, i.e. $G_{ab} = 0$ which causes the last two terms on the right hand side of Eq. (6.28) to vanish. Again writing $\phi_{\max}$ for the maximum field amplitude of its support over γ this gives

$$\int_{\gamma} d\lambda T_{ab} \dot{\gamma}^a \dot{\gamma}^b f^2 \geq -2\xi \phi_{\max}^2 \int_{\gamma} d\lambda (\partial_{\lambda} f)^2. \quad (6.33)$$

From Eq. (6.33) we can see that the amount of energy condition violation is determined by the maximum field amplitude. Now in the same way as above we can consider how Eq. (6.33) changes when we introduce the scaled smear function f_{λ_0} defined as

$$f_{\lambda_0}(\lambda) = \frac{f(\lambda/\lambda_0)}{\sqrt{\lambda_0}}, \quad \lambda_0 > 0, \quad (6.34)$$

with $\|f_{\lambda_0}\|^2 = \|f\|^2 = 1$ where $\|\cdot\|^2$ is the L^2 norm. We will also introduce the constant C_f as

$$C_f = \int_{\gamma} d\lambda (\partial_{\lambda} f)^2, \quad (6.35)$$

which is a constant since f is fixed. With this we get

$$\int_{\gamma} d\lambda T_{ab} \dot{\gamma}^a \dot{\gamma}^b f_{\lambda_0}^2 \geq -2\xi \phi_{\max}^2 C_f \cdot \frac{1}{\lambda_0^2}. \quad (6.36)$$

As $\lambda_0 \rightarrow 0$, f_{λ_0} approaches a delta function, while the right hand side of Eq. (6.36) goes to $-\infty$ (when $\xi > 0$) which highlights how the pointwise weak energy condition can be arbitrarily badly broken. When $\lambda_0 \rightarrow \infty$ we have that

$$\liminf_{\lambda_0 \rightarrow \infty} \int_{\gamma} d\lambda T_{ab} \dot{\gamma}^a \dot{\gamma}^b f_{\lambda_0}^2 \geq 0. \quad (6.37)$$

Now if $T_{ab} \dot{\gamma}^a \dot{\gamma}^b$ is integrable, using the dominated convergence theorem we have the following theorem, seen as theorem II.2 in [14].

Theorem 4. Let γ be a casual geodesic with affine parameter λ in a Ricci-flat spacetime (M, g) . Let T_{ab} be the stress tensor for the non-minimally coupled scalar field with coupling constant $\xi \in [0, 1/4]$. Then we have

$$\int_{\gamma} d\lambda T_{ab} \dot{\gamma}^a \dot{\gamma}^b \geq 0, \quad (6.38)$$

provided that the left hand side exists.

This gives us that when γ is null (timelike) Eq. (6.38) gives us the ANEC (AWEC) with Eq. (6.37) being a generalisation of these facts. Another approach, also explored above is to estimate the maximum value of $T_{ab} \dot{\gamma}^a \dot{\gamma}^b$ over an open interval I . Starting with Eq. (6.33) taking a supremum on the left hand side gives

$$\sup_{\gamma(I)} T_{ab} \dot{\gamma}^a \dot{\gamma}^b \geq -2\xi \phi_{\max}^2 \frac{\int_{\gamma(I)} d\lambda (\partial_{\lambda} f)^2}{\int_{\gamma(I)} d\lambda f^2}, \quad (6.39)$$

to extremising over all f on the right hand side gives using Eq. (6.25)

$$\sup_{\gamma(I)} T_{ab} \dot{\gamma}^a \dot{\gamma}^b \geq -2\xi \phi_{\max}^2 \frac{\pi^2}{\Gamma^2}, \quad (6.40)$$

where Γ is the length of the interval I . We see that long-lasting negative energy densities are attributed to large field amplitudes.

7 Quantum Energy Inequalities

It has long been known that in general quantum field theory violates all the classical pointwise energy conditions and some of the classical average ones given above. For example, in the WEC has the interpretation the energy density is non-negative, while in quantum field theories the equivalent quantity, the expectation of the energy density can be unbounded below. Despite this, we can still obtain conditions on a quantum field theory that have some resemblance to the classical energy conditions. The conditions

are called *quantum energy inequalities* (QEIs). We note the choice of the word “inequality” rather than “condition” as often (classical) energy conditions are imposed on a (classical) theory for the purposed of proving results, (e.g. singularity theorems), while often the (quantum) inequalities are derived from a quantum theory. In this section, we give some examples of simple QFTs violating classical energy conditions and the most common QEIs and their consequences. By doing this we provide a fuller picture of the current energy condition landscape in physics.

7.1 QFT violating energy conditions - example

The following example highlights how the simplest QFT can easily violate the classical energy conditions. We take the quantised real scalar field in four dimensional Minkowski space. We will see that the ‘Wick square’ of a field - the quantum analogue of the square of a classical field can take negative expectation values. Note in this section that our signature is $(+---)$. The standard quantisation is

$$\phi(x) = \int \frac{d^3\mathbf{k}}{(2\pi)^3\sqrt{2\omega}} (e^{-ikx}a(\mathbf{k}) + e^{ikx}a^*(\mathbf{k})), \quad (7.1)$$

where k is a 4-vector with $k^a = (\omega, \mathbf{k})$, kx denotes the Minkowski inner product, $\omega = \sqrt{|\mathbf{k}|^2 + m^2}$ and $a(\mathbf{k})$ and $a^*(\mathbf{k})$ are respectively annihilation and creation operators satisfying the commutation relations

$$[a(\mathbf{k}), a^*(\mathbf{k}')] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}') \mathbb{I}, \quad (7.2)$$

and also with $a(\mathbf{k})$ annihilating the vacuum state Ω for every $\mathbf{k}$, i.e.

$$a(\mathbf{k})\Omega = 0 \quad \forall \mathbf{k}. \quad (7.3)$$

We define the Wick square operator by $:\phi^2(x):$ where $::$ denotes normal ordering, and the smeared Wick square operator by

$$:\phi^2:(f) = \int d^4x : \phi^2(x) : f(x), \quad (7.4)$$

where $f \in C_0^\infty(\mathbb{R}^4)$, i.e. smooth function on $\mathbb{R}^4$ vanishing at infinity. We have the calculation

$$:\phi^2:(f)\Omega = \int \frac{d^3\mathbf{k}}{(2\pi)^3\sqrt{2\omega}} \frac{d^3\mathbf{k}'}{(2\pi)^3\sqrt{2\omega'}} \tilde{f}(k+k') a^*(\mathbf{k}) a^*(\mathbf{k}') \Omega, \quad (7.5)$$

where $\tilde{f}$ denotes the Fourier transform of f ,

$$\tilde{f}(k) = \int d^n x e^{ikx} f(x). \quad (7.6)$$

This immediately gives

$$\langle \Omega | : \phi^2 : (f) | \Omega \rangle = 0, \quad (7.7)$$

i.e. the expectation value of $:\phi^2:(f)$ at the vacuum vanishes. However, we can also calculate that

$$\| : \phi^2 : (f) \Omega \|^2 = \int \frac{d^3\mathbf{k}}{(2\pi)^3\sqrt{2\omega}} \frac{d^3\mathbf{k}'}{(2\pi)^3\sqrt{2\omega'}} \frac{|\tilde{f}(k+k')|^2}{2\omega\omega'}, \quad (7.8)$$

which is positive unless f is identically zero. So the expectation value of $:\phi^2:(f)$ in state Ω vanished while $:\phi^2:(f)$ does not annihilate Ω , which tells us that there exists some negative energy spectrum for $:\phi^2:(f)$. We choose f so that $\| : \phi^2 : (f) \Omega \| = 1$ and define ψ_α by

$$\psi_\alpha = \cos(\alpha)\Omega + \sin(\alpha) : \phi^2 : (f) \Omega, \quad (7.9)$$

then a calculation gives

$$\begin{aligned} \langle \psi_\alpha | : \phi^2 : (f) | \psi_\alpha \rangle &= \sin(2\alpha) + \sin^2(\alpha) \langle \Omega | : \phi^2 : (f)^3 | \Omega \rangle \\ &= 2\alpha + O(\alpha^2), \end{aligned} \quad (7.10)$$

and so for small $\alpha < 0$ the expectation of ψ_α in the state Ω can be negative. It can be further shown that the expectation of $:\phi^2:(f)$ in a suitable normalised state ψ can become unbounded from below, see [34].

7.2 Klein-Gordon field QEI

One of the simplest examples of a QEI is the massive Klein-Gordon field in 4-dimensional Minkowski space:

$$\int_{\gamma} \langle : T_{00} : \rangle_{\omega} g(t)^2 dt \geq -\frac{1}{16\pi^2} \int_{-\infty}^{\infty} |g''(t)|^2 dt, \quad (7.11)$$

where the integral is evaluated over an inertial observer world-line γ , and $\langle : T_{00} : \rangle_{\omega}$ is the expectation value of the renormalised energy density evaluated at a suitable state ω . Usually we take ω as *Hadamard states*, of which a precise definition can be found in [35], but for our purposes we can think of them as a large class of physically reasonable states. We also require that g is a smooth compactly supported function. The following are some remarks about this inequality from [36] are as follows. The left side contains the state ω , while the right side does not. It is not known if the bound is optimal. The smoothness of g is not strictly required - it is also valid in four dimensions if g is in the Sobolev space $W^{2,2}(\mathbb{R})$. We gain insight into this QEI by considering the scaling function $g_{\tau}(t) = \tau^{-1/2}g(t/\tau)$ so that Eq. (7.11) becomes

$$\frac{1}{\tau} \int_{\gamma} \langle : T_{00} : \rangle g(t/\tau)^2 dt \geq -\frac{1}{16\pi^2} \frac{1}{\tau^4} \int_{-\infty}^{\infty} |g''(t)|^2 dt. \quad (7.12)$$

The limit $\tau \rightarrow 0^+$ is viewed as the short-time sampling limit and so in Eq. (7.12) confirms that the energy density expectation is unbounded below. In the limit $\tau \rightarrow \infty$ we have

$$\liminf_{\tau \rightarrow \infty} \int_{\gamma} \langle : T_{00} : \rangle g(t/\tau)^2 dt \geq 0, \quad (7.13)$$

and so the AWEC holds, while the WEC does not. As discussed in Sec. 5, this gives further motivation for using averaged energy conditions over pointwise energy conditions.

8 Weakening energy conditions in the Cosmic No-Hair Theorem

Our next aim is to try and weaken the energy conditions used in Wald's Theorem, namely the DEC and the SEC and see what kind of results we obtain. The approach we take is to assume the SEC, but take an averaged weaker version of the WEC, and a pointwise weakened version of the DEC. Let us assume we have an averaged energy condition given by

$$\int_0^{\infty} d\tau f^2(\tau) T_{ab} n^a n^b \geq -A, \quad (8.1)$$

where $A \geq 0$, and the integral is performed over a half timelike geodesic $\gamma : [0, \infty) \rightarrow M$ with tangent vector n^a . We will discuss potential conditions that may have to be placed on f shortly. Now from Eq. (3.48), we see that multiplying by f^2 and integrating along γ gives

$$\int_0^{\infty} d\tau f^2(\tau) (K^2 - 3\Lambda) \geq 24\pi \int_0^{\infty} d\tau f^2(\tau) T_{ab} n^a n^b. \quad (8.2)$$

Now assuming the SEC, we know from Sec. 4 that

$$\frac{3\Lambda}{\sinh^2(\tau/\alpha)} \geq (K^2 - 3\Lambda), \quad (8.3)$$

and so multiplying by f^2 , and also using our weakened weak average energy condition Eq. (8.1), we have the inequality

$$\int_0^{\infty} d\tau f^2(\tau) \frac{3\Lambda}{\sinh^2(\tau/\alpha)} \geq \int_0^{\infty} d\tau f^2(\tau) (K^2 - 3\Lambda) \geq 24\pi \int_0^{\infty} d\tau f^2(\tau) T_{ab} n^a n^b \geq -24\pi A. \quad (8.4)$$

With Eq. (8.4) in mind, our aim now is to choose the right conditions on f such that the second integral in Eq. (8.4) not only converges, but so that K^2 is forced to approach 3Λ . For example, we need an f so that the zeros of f^2 at $\tau = 0$ are higher order than the poles of the $\sinh^2$ at $\tau = 0$. Such functions could be

$$f(\tau) = \operatorname{erf}(\tau) \quad \text{or} \quad f(\tau) = \left(\frac{2}{1 + e^{-x}} - 1 \right), \quad (8.5)$$

or similar sigmoid functions. These functions also have the property that $f^2(\tau) \rightarrow 1$ as $\tau \rightarrow \infty$ which would ensure that the left-most integral in Eq. (8.4) converges. The following theorem establishes conditions on f , and in fact also on K , so that $K^2 \rightarrow 3\Lambda$ at late times.

Theorem 5. Suppose $f \in C^1(\mathbb{R})$ is chosen so that the integral on the left-hand side of Eq. (8.4) converges and $\lim_{\tau \rightarrow \infty} f^2(\tau) = 1$. Suppose also that K^2 is uniformly continuous. Then $K^2 \rightarrow 3\Lambda$.

Proof. Let $g(\tau) = K^2 - 3\Lambda$, so g is uniformly continuous. Aiming for a contradiction, assume that

$$\lim_{\tau \rightarrow \infty} g(\tau) \neq 0, \quad (8.6)$$

then, we can pick a $c > 0$ and a sequence $\{\tau_n\}$ such that

$$\tau_n \geq \tau_{n-1} + 1 \quad \text{and} \quad |g(\tau_n)| > c. \quad (8.7)$$

Also, since $\lim_{\tau \rightarrow \infty} f^2(\tau) = 1$, unpacking the definition of the limit, we have a T such that

$$|f^2(\tau) - 1| < \frac{1}{2}, \quad \forall \tau > T, \quad (8.8)$$

which in particular means

$$f^2(\tau) > \frac{1}{2}, \quad \forall \tau > T. \quad (8.9)$$

Now, as g is uniformly continuous, there exists $0 < \delta < 1$ such that

$$|g(\tau) - g(\tau')| < \frac{c}{2}, \quad \text{whenever} \quad |\tau - \tau'| < \delta. \quad (8.10)$$

Now, consider for each n the interval $I_n = [\tau_n - \delta/2, \tau_n + \delta/2]$, so that we have $g(\tau)|_{I_n} \geq c/2$. We can also pick an N large enough so that $\tau_N > T$, and so $f^2(\tau) > 1/2$ for all $\tau > \tau_N$. Now, we have the result that

$$\left| \int_{I_n} f^2(\tau) g(\tau) d\tau \right| \geq \frac{1}{2} \cdot \frac{c}{2} \cdot \delta, \quad (8.11)$$

for all $n > N$. Since the intervals I_n are disjoint, the integral

$$\int_0^\infty d\tau f^2(\tau) (K^2 - 3\Lambda) \quad (8.12)$$

in Eq. (8.4) doesn't converge and we have a contradiction. Therefore,

$$\lim_{\tau \rightarrow \infty} g(\tau) = 0 \quad \text{or equivalently} \quad K^2 \rightarrow 3\Lambda. \quad (8.13)$$

□

Before continuing, we need to discuss whether the assumptions on f and K used to prove Theorem 5 are reasonable. The assumption that K^2 is uniformly continuous is used to disallow behaviour of the sort of increasing tent functions or increasing oscillation functions that may result in the integral Eq. (8.12) converging without K^2 approaching 3Λ . However, only using the SEC, we know that K^2 is pointwise bounded. From Eq. (4.7) we can obtain

$$0 \leq K^2 \leq \frac{(3\Lambda)}{\tanh^2(\tau/\alpha)}, \quad (8.14)$$

and so at late times,

$$0 \leq K^2 \leq 3\Lambda. \quad (8.15)$$

This argument excludes K^2 from behaving like increasing tent functions, e.g., the continuous function that connects the points $(0, 0)$, $(n - 1/n^3, 0)$, (n, n) and $(n + 1/n^3, 0)$ with straight line segments for $n \geq 2$. We should note that uniform continuity of K^2 excludes these types of functions anyway, but the fact that K^2 is bounded confirms that this type of behaviour is not allowed. In regards to K^2 having increasing-oscillation behaviour, e.g. $\sin(\tau^2)$, it can be argued that such behaviour would be unphysical in our universe. Indeed in this case the derivative of K^2 becomes unbounded and it could be seen as reasonable to neglect these cases where the universe has an unbounded acceleration of expansion.

The conditions placed on f in Theorem 5 are more questionable. The condition that the left-hand integral of Eq. (8.4) converges is done with the specific purpose of being able to prove Theorem 5 and could be seen as a case of “working backwards to prove a certain result”. The condition that $f^2(\tau) \rightarrow 1$ as $\tau \rightarrow \infty$ could also be seen as an example of this, and is different to the requirements of the theorems in Sec. 6, where we require that $f(\tau)$ vanishes as $\tau \rightarrow \infty$ has compact support. This being said, although the limit of 1 is arbitrary (any positive constant achieves the same result), choosing 1 enables that at late times the integrand in the energy condition Eq. (8.1) tends towards $T_{ab}n^an^b$ and so Eq. (8.1) looks like the standard half-AWEC (since γ starts at 0) with an improved lower bound.

One consequence of Theorem 5 is that it implies that at late times, the WEC is indeed violated (or if not violated then $T_{ab}n^an^b$ is identically zero). We can see this by considering the inequality that comes from Eq. (3.48),

$$K^2 - 3\Lambda \geq 24\pi T_{ab}n^an^b. \quad (8.16)$$

Since at late times $K^2 \rightarrow 3\Lambda$, this means

$$T_{ab}n^an^b \leq 0 \quad (8.17)$$

at late times. So this weaker energy condition we have assumed doesn't just present the possibility the WEC is violated, it actually ensures that the WEC is violated. In general, any energy condition that ensures $K^2 \rightarrow 3\Lambda$ without placing any direct condition on $T_{ab}n^an^b$ itself, will ensure that the WEC is violated at late times.

We can pursue our line of argument slightly further. From Eq. (3.48), again we multiply by f^2 and integrate along γ , but now we don't absorb $\sigma_{ab}\sigma^{ab}$ into the inequality. Doing this, we obtain

$$\begin{aligned} \int_0^\infty d\tau f^2(\tau)(K^2 - 3\Lambda) &\geq \frac{3}{2} \int_0^\infty d\tau f^2(\tau)\sigma_{ab}\sigma^{ab} + 24\pi \int_0^\infty d\tau f^2(\tau)T_{ab}n^an^b \\ &\geq \frac{3}{2} \int_0^\infty d\tau f^2(\tau)\sigma_{ab}\sigma^{ab} - 24\pi A, \end{aligned} \quad (8.18)$$

where to obtain the second line we use Eq. (8.1). Now, since the left-hand side integral of Eq. (8.18) is bounded, and since that $\sigma_{ab}\sigma^{ab} \geq 0$, we can conclude that the integral

$$\int_0^\infty d\tau f^2(\tau)\sigma_{ab}\sigma^{ab} \quad (8.19)$$

converges. Via the same argument as Theorem 5, if we assume that $\sigma_{ab}\sigma^{ab}$ is uniformly continuous (viewed as a function of τ along the geodesic), then we must have $\sigma_{ab}\sigma^{ab} \rightarrow 0$ as $\tau \rightarrow \infty$. This assumption comes with the same discussion above regarding the same assumption placed on K^2 . Since σ_{ab} has spatial indices, as discussed in Sec. 4 this means that we have $\sigma_{ab} \rightarrow 0$ as $\tau \rightarrow \infty$.

We can use the same argument a final time: Eq. (8.4) tells us that

$$\int_0^\infty d\tau f^2(\tau)T_{ab}n^an^b \quad (8.20)$$

converges, and so if $T_{ab}n^an^b$ is uniformly continuous, then at late times $T_{ab}n^an^b \rightarrow 0$, i.e. the energy density ρ measured by an observer travelling along γ approaches zero. At the corresponding point in Sec. 4, we employed the DEC to show that if the energy density approaches zero, then the components of the stress tensor T_{ab} individually approach zero. This method is not possible here, since assuming the DEC would imply the WEC which we aimed to weaken (and is in fact violated). Nonetheless, since we already have $\rho \rightarrow 0$ at late times, if we assume a slightly weakened version of the second part of the effective description of the DEC, $|\rho| \geq |P_i|$, then this is enough to force each component of the stress tensor $T_{ab} \rightarrow 0$. Note that we cannot simply take the condition $\rho \geq |P_i|$ because we know that ρ can be negative.

The final part of the argument is the same as in Sec. 4. Since $K^2 \rightarrow 3\Lambda$, then either $K \rightarrow (3\Lambda)^{1/2}$ or $K \rightarrow -(3\Lambda)^{1/2}$, in which case the spatial metric is approximated at late times by

$$h_{ab}(\tau) = e^{\pm 2\tau/\alpha} h_{ab}(0), \quad (8.21)$$

where the $+$ solution corresponds to $K \rightarrow (3\Lambda)^{1/2}$ and the $-$ solution corresponds to $K \rightarrow -(3\Lambda)^{1/2}$. Now, as in Sec. 4, we can conclude that the universe locally approaches de Sitter spacetime.

The main issue with this argument is that the f we choose, as discussed, does not possess the usual requirements of compact support and vanishing at infinity. Here, we do not attempt to completely remedy this problem, but we can suggest a method from [19] which may lead to similar results without the need to impose heavy restrictions on f .

To begin, we could choose a weakened energy condition of the sort of

$$\int_{\gamma} d\tau T_{ab}n^an^bg^2(\tau) \geq -Q_m(\gamma)\|g^{(m)}\|^2 - Q_0(\gamma)\|g\|^2, \quad (8.22)$$

where $\|\cdot\|$ denotes the L^2 norm, and Q_m and Q_0 are constants that depend on the curve γ . Following the same steps as the argument above, we obtain the inequality

$$-Q_m(\gamma)\|g^{(m)}\|^2 - Q_0(\gamma)\|g\|^2 \leq \int_{\gamma} d\tau (K^2 - 3\Lambda)g^2(\tau) \leq 3\Lambda \int_{\gamma} d\tau \frac{g^2(\tau)}{\sinh^2(\tau/\alpha)}. \quad (8.23)$$

Now, the main idea is to take two functions, $f, \varphi \in C^\infty(\mathbb{R})$ and a point $\tau_0 \in (0, \infty)$ such that $f(0) = 0$, $f(\tau) = 1$ for $\tau \in [\tau_0, \infty]$ and $\varphi(\infty) = 0$ and $\varphi = 1$ for $\tau \in [0, \tau_0]$. Then, we have the functional equality

$$f^2 = (f\varphi)^2 + 1 - \varphi^2, \quad (8.24)$$

which we substitute into Eq. (8.23) after letting $g = f\varphi$. This is now useful because we have the equality

$$\int_0^\infty d\tau g^2 = \int_0^{\tau_0} d\tau f^2 + \int_{\tau_0}^\infty d\tau \varphi^2, \quad (8.25)$$

which divides the integral over γ into separate integrals over $(0, \tau_0)$ and (τ_0, ∞) . This is now useful, since we have $f(0) = 0$ which (assuming the zero is high enough order), will control the behaviour of $\sinh^2(\tau/\alpha)$ at zero on the right-hand side of Eq. (8.23). We may hope that we are able to use this to bound the integral, and then proceed with similar arguments to those above, however now, would be continuing, by construction, with a compactly supported g .

9 Conclusion

In this thesis, we have given an overview of the main types of energy conditions used in physics today. We have seen how initial results using pointwise energy conditions were successful in proving many

theorems about the universe, but their usage comes with many drawbacks and questions about their general validity. We then saw how averaged conditions were used to attempt to remedy the problems of pointwise conditions. Their importance was threefold: first, they provided energy conditions that were harder to violate than their pointwise counterparts. They achieve this whilst still capturing the essential notions that energy conditions were created to demonstrate, like the positivity of energy and the attractiveness of gravity. Secondly, they were used to prove new theorems about the structure of the universe, like the impossibility of wormholes and closed timelike curves. Thirdly, on a related note, they had success in generalising theorems that were originally proven solely using pointwise energy conditions. This third point represents the main aim of this thesis. We discussed in detail Wald's Cosmic No-Hair Theorem, which uses solely pointwise energy conditions in its results, and were then able to prove an analogous result with weaker averaged energy conditions. In this process however, we saw that often stronger conditions on the structure of the spacetime have to be placed in order for certain arguments to hold. Nevertheless, this is not to say that these exact conditions are required to be present, and there is certainly room for more investigation. There are many other techniques and even other types of weaker energy conditions that could be used to prove results similar to the ones discussed here. It is clear that the broad topic of energy conditions in general relativity will continue to have a principle role in the study and understanding of our universe.

Appendices

A Calculation of Eq. (3.59)

For clarity, let us break up the calculation of ${}^{(3)}R$ into

$${}^{(3)}R = \underbrace{-C_{ab}^a C_c^b}_{(1)} + \underbrace{\frac{1}{2}C_{bc}^a C_{ab}^c}_{(2)} - \underbrace{\frac{1}{4}C_{abc} C^{abc}}_{(3)}. \quad (\text{A.1})$$

Considering (1) we have

$$\begin{aligned} -C_{ab}^a C_c^b &= -C_{ab}^a h^{bd} C_{cd}^c \\ &= -h^{bd} A_b A_d \\ &= -A_b A^b, \end{aligned} \quad (\text{A.2})$$

recalling the definition of A_a in Eq. (3.50). Considering (2), we have

$$\begin{aligned} \frac{1}{2}C_{bc}^a C_a^b &= \frac{1}{2}C_{bc}^a h^{bd} C_{ad}^c \\ &= \frac{1}{2}h^{bd}(M^{ae}\epsilon_{ebc} + \delta_{[b}^a A_{c]})(M^{cf}\epsilon_{fad} + \delta_{[a}^c A_{d]}) \\ &= \frac{1}{2}h^{bd}\left[\epsilon_{ebc}\epsilon_{fad}M^{ae}M^{cf} + \frac{1}{2}\epsilon_{fad}M^{cf}(\delta_{[b}^a A_{c]} - \delta_{[c}^a A_{b]})\right. \\ &\quad \left.+ \frac{1}{2}\epsilon_{ebc}M^{ae}(\delta_{[a}^c A_{d]} - \delta_{[d}^c A_{a]}) + \frac{1}{4}(\delta_{[b}^a A_{c]} - \delta_{[c}^a A_{b]})(\delta_{[a}^c A_{d]} - \delta_{[d}^c A_{a]})\right] \\ &= \frac{1}{2}h^{bd}\left[\epsilon_{ebc}\epsilon_{fad}M^{ae}M^{cf} - \frac{1}{2}\epsilon_{fad}M^{af}A_b + \frac{1}{2}\epsilon_{ebc}M^{ce}A^d\right. \\ &\quad \left.+ \frac{1}{4}(\delta_{[b}^c A_{c]}A_d - \delta_{[a}^a A_b A_d - A_b A_d + \delta_{[d}^a A_b A_a])\right] \\ &= \underbrace{\frac{1}{2}h^{bd}\epsilon_{ebc}\epsilon_{fad}M^{ae}M^{cf}}_{(A)} - \frac{1}{4}A_b A^b, \end{aligned} \quad (\text{A.3})$$

where we have used Eq. (3.56) as well as $\epsilon_{abc}M^{ab} = 0$. We will leave the term marked (A) for now and continue with (3). We have

$$\begin{aligned} -\frac{1}{4}C_{abc}C^{abc} &= -\frac{1}{4}h^{ad}C_{bc}^d h^{be}h^{cf}C_{ef}^a \\ &= -\frac{1}{4}h_{ad}h^{be}h^{cf}(M^{dg}\epsilon_{gbc} + \delta_{[b}^d A_{c]})(M^{ah}\epsilon_{hef} + \delta_{[e}^a A_{f]}) \\ &= -\frac{1}{4}h_{ad}h^{be}h^{cf}\left[\epsilon_{gbc}\epsilon_{hef}M^{dg}M^{ah} + \frac{1}{2}\epsilon_{hef}M^{ah}(\delta_{[b}^d A_{c]} - \delta_{[c}^d A_{b]})\right. \\ &\quad \left.+ \frac{1}{2}\epsilon_{gbc}M^{dg}(\delta_{[e}^a A_{f]} - \delta_{[f}^a A_{e]}) + \frac{1}{4}(\delta_{[b}^d A_{c]} - \delta_{[c}^d A_{b]})(\delta_{[e}^a A_{f]} - \delta_{[f}^a A_{e]})\right] \\ &= -\frac{1}{4}h_{ab}h^{be}h^{cf}\epsilon_{gbc}\epsilon_{hef}M^{dg}M^{ah} - \frac{1}{8}\epsilon_{hef}M^{ah}(\delta_{[a}^e h^{cf}A_{c]} - \delta_{[c}^f h^{be}A_{b]}) \\ &\quad - \frac{1}{8}\epsilon_{gbc}M^{dg}(\delta_{[d}^b h^{cf}A_{f]} - \delta_{[c}^d h^{be}A_{e]}) \\ &\quad - \frac{1}{16}(\delta_{[a}^e h^{cf}A_{c]} - \delta_{[c}^f h^{be}A_{b]})(\delta_{[e}^a A_{f]} - \delta_{[f}^a A_{e]}) \\ &= -\frac{1}{4}h_{ab}h^{be}h^{cf}\epsilon_{gbc}\epsilon_{hef}M^{dg}M^{ah} - \frac{1}{16}(3A_b A^b - A_b A^b - A_b A^b + 3A_b A^b) \\ &= \underbrace{-\frac{1}{4}h_{ab}h^{be}h^{cf}\epsilon_{gbc}\epsilon_{hef}M^{dg}M^{ah}}_{(B)} - \frac{1}{4}A_b A^b, \end{aligned} \quad (\text{A.4})$$

where in line four we have used $\epsilon_{abc}M^{ab} = 0$ after applying the Kronecker deltas on the M_{ab} terms. To simplify the terms (A) and (B) we use the following expression for the determinant of h^{ab} :

$$h^{-1}\epsilon^{abc} = \epsilon_{def}h^{ad}h^{be}h^{cf}, \quad (\text{A.5})$$

where $h^{-1} = \det(h^{ab})$. Considering (A), we have

$$\begin{aligned} \frac{1}{2}h^{bd}\epsilon_{ebc}\epsilon_{fad}M^{ae}M^{cf} &= \frac{1}{2}h^{bd}h^{eg}h^{ch}\epsilon_{ebc}\epsilon_{fad}M_g^a M_h^f \\ &= \frac{1}{2}h^{-1}\epsilon^{gdh}\epsilon_{fad}M_g^a M_h^f \\ &= \frac{1}{2}h^{-1}(\delta_f^h\delta_a^g - \delta_a^h\delta_f^g)M_g^a M_h^f \\ &= \frac{1}{2}h^{-1}(M_h^h M_a^a - M_g^h M_h^g) \\ &= \frac{1}{2}h^{-1}(M^2 - M_{ab}M^{ab}). \end{aligned} \quad (\text{A.6})$$

Considering (B), we have

$$\begin{aligned} -\frac{1}{4}h_{ab}h^{be}h^{cf}\epsilon_{gbc}\epsilon_{hef}M^{dg}M^{ah} &= -\frac{1}{4}h^{gd}h^{be}h^{cf}\epsilon_{gbc}\epsilon_{hef}M_{ad}M^{ah} \\ &= -\frac{1}{4}h^{-1}\epsilon^{def}\epsilon_{hef}M_{ad}M^{ah} \\ &= -\frac{1}{4}h^{-1} \cdot 2\delta_h^d M_{ad}M^{ah} \\ &= -\frac{1}{2}h^{-1}M_{ab}M^{ab}. \end{aligned} \quad (\text{A.7})$$

Putting together everything gives

$$\begin{aligned} {}^{(3)}R &= -A_b A^b + \frac{1}{2}h^{-1}(M^2 - M_{ab}M^{ab}) \\ &\quad -\frac{1}{4}A_b A^b - \frac{1}{2}h^{-1}M_{ab}M^{ab} - \frac{1}{4}A_b A^b \\ &= -\frac{3}{2}A_b A^b - h^{-1}(M_{ab}M^{ab} - \frac{1}{2}M^2), \end{aligned} \quad (\text{A.8})$$

which is what we aimed to calculate.

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